

PartCo: Part-Level Correspondence Priors Enhance Category Discovery

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<https://visual-ai.github.io/partco>

Abstract

Generalized Category Discovery (GCD) aims to identify both known and novel categories within unlabeled data by leveraging a set of labeled examples from known categories. Existing GCD methods primarily depend on semantic labels and global image representations, often overlooking the detailed part-level cues that are crucial for distinguishing closely related categories. In this paper, we introduce PartCo, short for Part-Level Correspondence Prior, a novel framework that enhances category discovery by incorporating part-level visual feature correspondences. By leveraging part-level relationships, PartCo captures finer-grained semantic structures, enabling a more nuanced understanding of category relationships. Importantly, PartCo seamlessly integrates with existing GCD methods without requiring significant modifications. Our extensive experiments on multiple benchmark datasets demonstrate that PartCo significantly improves the performance of current GCD approaches, outperforming most existing methods by bridging the gap between semantic labels and part-level visual compositions, thereby setting new benchmarks for GCD.

1 Introduction

Supervised deep learning models have fundamentally transformed computer vision, showcasing exceptional proficiency in classifying predefined image categories. Models trained on extensive labeled datasets achieve high accuracy and robustness in distinguishing known classes within controlled environments. However, their performance significantly diminishes when confronted with samples from categories that were neither present nor represented dur-

ing training. This limitation impedes the deployment of intelligent systems in dynamic, real-world scenarios where encountering previously unseen categories is inevitable. To address this challenge, Generalized Category Discovery (GCD) (Vaze et al., 2022a) has emerged as a pivotal task. As depicted in Fig. 1, GCD aims to automatically identify and categorize both known and novel classes within unlabeled data by leveraging a modest set of labeled examples from known categories. Unlike traditional supervised learning, which operates within a rigid framework of predefined categories, GCD extends the model’s capability to recognize and incorporate unseen categories alongside known ones.

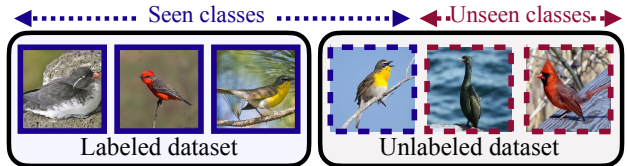


Figure 1. **Generalized Category Discovery:** Given a labeled subset contains seen classes, the task is to categorize the unlabeled images, which may belong to seen or unseen classes.

A growing body of literature in GCD (He et al., 2025a), emphasizes the significance of object parts as effective conduits for transferring knowledge between “seen” and “unseen” categories (Vaze et al., 2022a; Wang et al., 2024). Object parts (Fig. 2) encapsulate fine-grained visual features whose correspondences are often shared across different categories, facilitating the generalization to novel classes.

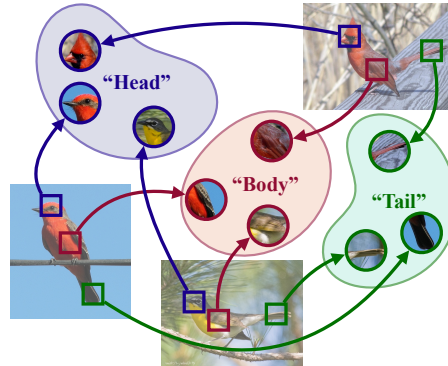


Figure 2. **Part variability.** Object parts (head, body, tail) vary in scale, pose, and visibility yet still correspond across images, motivating part-aware priors beyond global features.

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However, recent approaches predominantly rely on global representations derived from the classification token of transformer-based models. While these global features capture the overall semantic content of an image, they inherently abstract away detailed part-level information, which is vital for distinguishing closely related categories. For instance, Wang et al. (2024) introduces a spatial prompt tuning method that learns pixel-level prompts around local image regions to incorporate part-level information. Although innovative, this method does not account for the inherent variability in object parts, such as differing scales, orientations, or varying numbers of parts due to occlusions. This motivates an explicit part-aware prior that is robust to scale, pose, and occlusion.

Vision Transformer (ViT) models, in addition to the classification token, incorporate patch tokens that encapsulate high-dimensional features for each image patch. These patch tokens inherently contain part-level observations, offering a granular perspective of the image’s composition. However, directly utilizing these patch tokens presents several challenges: the absence of explicit part-level information, the presence of foreground-background noise, and varying object scales and orientations across samples. These issues necessitate an effective *supervisory signal* to fully harness the potential of patch token representations in ViT models.

Recent advancements in self-supervised vision foundation models, particularly the ViT-based DINO variants (Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025), have demonstrated remarkable generalizability across various tasks. These models excel at extracting high-dimensional patch token features that capture detailed and localized semantic information for each image patch. Unlike the classification token, which summarizes the entire image, patch tokens focus on specific parts, providing a more detailed view of the image’s composition. Importantly, these enhanced feature descriptors inherently provide the necessary part-level correspondence labels, serving as an ideal supervisory signal for leveraging patch tokens within the framework.

To address these challenges, we propose **PartCo**, short for **Part-Level Correspondence Prior**, a versatile framework designed to introduce part-level correspondence labels into the GCD process. By explicitly guiding ViT patch token features with these correspondence labels, PartCo better leverages the utilization of the model’s rich feature representations. Additionally, we introduce a novel part-level correspondence loss that effectively leverages these part-level features, ensuring that detailed object part information is accurately captured and utilized for category discovery. Through comprehensive evaluations on both fine-grained and generic benchmark datasets, PartCo significantly improves GCD methods and outperforms most existing methods, setting a new standard for the GCD task.

2 Preliminaries

Problem statement. Generalized Category Discovery (GCD) aims to develop a model that accurately classifies unlabeled samples from known categories while simultaneously clustering those from novel, unseen categories. Consider an unlabeled dataset $\mathbf{D}_u = \{(\mathbf{x}_i^u, y_i^u)\} \subset \mathbf{X} \times \mathbf{Y}_u$ and a labeled dataset $\mathbf{D}_l = \{(\mathbf{x}_i^l, y_i^l)\} \subset \mathbf{X} \times \mathbf{Y}_l$, where $\mathbf{Y}_u$ and $\mathbf{Y}_l$ represent the label sets for unlabeled and labeled data, respectively. The unlabeled dataset contains samples from both known categories (included in $\mathbf{Y}_l$) and unknown categories, specifically $\mathbf{Y}_l \subset \mathbf{Y}_u$. Let $M = |\mathbf{Y}_l|$ denote the number of labeled categories. We assume the total number of categories, $K = |\mathbf{Y}_l \cup \mathbf{Y}_u|$, is known, as established in prior studies (Han et al., 2021; Vaze et al., 2023). When this information is unavailable, methods such as those in (Han et al., 2019; Vaze et al., 2022a; Hao et al., 2024; Zhao et al., 2023) can provide reliable estimates.

Baselines. The *non-parametric* baseline, (e.g., Vaze et al., 2022a) is introduced by fine-tuning the pretrained DINO model (Caron et al., 2021). The loss function generally integrates both self-supervised and supervised contrastive losses. For two augmented views $\mathbf{x}_i$ and $\mathbf{x}_i'$ over a mini-batch B , we obtain ℓ_2 -normalized features $\mathbf{z}_i = \psi(\phi(\mathbf{x}_i))$ and $\mathbf{z}_i' = \psi(\phi(\mathbf{x}_i'))$, where ϕ is the backbone and ψ is the projection head; τ_r denotes the temperature parameter. The contrastive losses are then defined as:

$$\begin{aligned}\mathcal{L}_{rep}^u &= \frac{1}{|B|} \sum_{i \in B} -\log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_i' / \tau_r)}{\sum_{j \neq i} \exp(\mathbf{z}_i \cdot \mathbf{z}_j' / \tau_r)}, \\ \mathcal{L}_{rep}^s &= \frac{1}{|B_l|} \sum_{i \in B_l} \frac{1}{|\mathbb{N}_i|} \sum_{q \in \mathbb{N}_i} -\log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_q / \tau_r)}{\sum_{j \neq i} \exp(\mathbf{z}_i \cdot \mathbf{z}_j / \tau_r)}.\end{aligned}\quad (1)$$

Here, $\mathbb{N}_i$ contains indices of labeled samples sharing the same label y_i^l as $\mathbf{x}_i$. The total representation loss $\mathcal{L}_{rep}$ is a weighted combination:

$$\mathcal{L}_{rep} = (1 - \lambda_b) \mathcal{L}_{rep}^u + \lambda_b \mathcal{L}_{rep}^s, \quad (2)$$

where λ_b is the balancing factor.

The *parametric* baseline, (e.g., Wen et al., 2023) employs a parametric classifier within a self-distillation framework (Caron et al., 2021). Initialized with K normalized category prototypes $\mathbf{L} = \{\mathbf{l}_1, \dots, \mathbf{l}_K\}$, the classifier computes the probability for category k as:

$$\mathbf{p}_i^{(k)} = \frac{\exp(\mathbf{o}_i \cdot \mathbf{l}_k / \tau_s)}{\sum_{j=1}^K \exp(\mathbf{o}_i \cdot \mathbf{l}_j / \tau_s)}, \quad (3)$$

where $\mathbf{o}_i = \phi(\mathbf{x}_i) / \|\phi(\mathbf{x}_i)\|$ and τ_s is the student temperature. Soft labels $\mathbf{q}_i$ are generated by a teacher network

with temperature τ_t . The unsupervised classification loss $\mathcal{L}_{cls}^u$ is defined as $\mathcal{L}_{cls}^u = \frac{1}{|B|} \sum_{i \in B} \ell_{ce}(\mathbf{q}_i', \mathbf{p}_i) - \xi \mathcal{H}(\bar{\mathbf{p}})$, where $\bar{\mathbf{p}} = \frac{1}{2|B|} \sum_{i \in B} (\mathbf{p}_i + \mathbf{p}_i')$ denotes the mean prediction across the mini-batch, ℓ_{ce} is the cross-entropy loss and $\mathcal{H}$ is the mean entropy, weighted by ξ . For labeled samples, the supervised loss $\mathcal{L}_{cls}^s = \frac{1}{|B_l|} \sum_{i \in B_l} \ell_{ce}(\mathbf{p}_i, \mathbf{y}_i)$ is used. The

overall classification loss combines unsupervised and supervised components as: $\mathcal{L}_{cls} = (1 - \lambda_b) \mathcal{L}_{cls}^u + \lambda_b \mathcal{L}_{cls}^s$. Finally, integrating with the non-parametric representation loss in Eq. 2 yields the comprehensive GCD objective:

$$\mathcal{L}_{gcd} = \mathcal{L}_{cls} + \mathcal{L}_{rep}. \quad (4)$$

Limitation of baselines. Although the non-parametric and parametric baselines obtain encouraging results on GCD, they exhibit significant limitations. Primarily, these methods rely solely on the foundation model’s classification token ($[\text{CLS}]$) representation, which captures only global information about the input data. This exclusive dependence on global representations restricts the models from leveraging part-level or localized information that is essential for distinguishing fine-grained categories. Without incorporating detailed, part-specific features, the baselines may overlook subtle patterns and contextual nuances within the data, leading to less effective performance in category discovery.

3 PartCo: Part-Level Correspondence Prior Framework

Building upon the motivations outlined in the introduction, we introduce *PartCo*, a novel framework meticulously crafted to harness part-level information from ViT’s patch tokens for GCD. Unlike traditional approaches that rely solely on global representations provided by the $[\text{CLS}]$ token, PartCo fully leverages the rich, high-dimensional features embedded within ViT’s patch tokens. By generating and utilizing explicit part-level correspondence labels, PartCo effectively bridges the gap between coarse global features and fine-grained local details. Furthermore, this design allows PartCo to be seamlessly integrated into existing GCD methods, enhancing their performance without necessitating significant modifications.

By making use of these part-level correspondence labels, PartCo fully utilizes the vision foundation model beyond just the $[\text{CLS}]$ token. These labels act as robust supervisory signals, guiding the patch token features to focus on meaningful object parts and mitigating common challenges such as foreground-background noise and variability in object scales and orientations. This guidance enables the full potential of vision foundation models to be realized, ensuring that both global and local feature representations are explicitly integrated into the GCD process. In the subsequent sections, we detail the construction and utilization of part-level

correspondence labels within the PartCo framework.

3.1 Constructing part-level correspondence labels

To construct part-level correspondence labels, we employ a two-step process, illustrated in Fig. 3, leveraging the rich feature representations from the frozen DINOv2 model. This approach ensures robust label inference across both labeled and unlabeled samples by first acquiring relevant PCA projections and then assigning labels through k -means.

Step 1: PCA projections. We begin by sampling a subset of M labeled images from the dataset $\mathbf{D}_l$, ensuring that each sample represents a distinct category. From these images, we extract their patch token features denoted as $\mathbf{F} \in \mathbb{R}^{M \times N \times d}$ using vision foundation model ϕ , e.g., DINO backbone (Oquab et al., 2024; Siméoni et al., 2025), where N is the number of patch tokens and d is the feature dimension. The first principal component analysis (PCA) is applied to $\mathbf{F}$ to obtain the primary projection vector $\mathbf{w}_{obj} \in \mathbb{R}^d$, which captures the most significant variation corresponding to object regions: $\mathbf{w}_{obj} = \arg\max_{\mathbf{w}} \frac{\mathbf{w}^\top \mathbf{F}^\top \mathbf{F} \mathbf{w}}{\mathbf{w}^\top \mathbf{w}}$. Using this projection, we compute the objectness score for each patch: $\mathbf{F}_{obj} = \mathbf{F} \cdot \mathbf{w}_{obj}$, and generate a binary mask $\mathbf{M}$ by thresholding at $\tau_{obj} = 0.6$: $\mathbf{M} = \mathbb{1}(\mathbf{F}_{obj} > \tau_{obj})$. This mask distinguishes foreground patches from background ones. Subsequently, we perform a second PCA on the masked features to extract fine-grained information. Specifically, we compute the element-wise multiplication $\mathbf{F} \odot \mathbf{M}$ and apply PCA to obtain the projection matrix $\mathbf{w}_{fg} \in \mathbb{R}^{d \times 3}$, resulting in the fine-grained feature representation: $\mathbf{F}_{fg} = (\mathbf{F} \odot \mathbf{M}) \cdot \mathbf{w}_{fg}$. This transformation maps the first three components of the PCA computed over the feature space to RGB.

Step 2: Label construction. We determine the optimal number of part-level labels, k^* , by applying k -means clustering to the normalized fine-grained features $\mathbf{F}_{fg}$. Each clustering solution is evaluated based on two criteria: (1) *minimum distance* between cluster centers to ensure that the clusters are well separated, reducing overlap and increasing distinctiveness. (2) *balance of cluster sizes*: prevents skewed distributions where some clusters dominate over others, promoting uniformity. We sweep k over a candidate set and select k^* by maximizing $\min_{i \neq j} \|\mathbf{c}_i - \mathbf{c}_j\| \times (\min_i |C_i| / \max_j |C_j|)$, favoring well-separated and balanced clusters. Here, $\mathbf{c}_i$ and $\mathbf{c}_j$ denote the centroids of clusters i and j , and $|C_i|$ is the number of samples in cluster i . With k^* fixed, we assign part-level correspondence labels to all samples in $\mathbf{D}$. We then define the part label map $y_i^{\text{part}} \in \{1, \dots, k^*\}$ with resolution following ViT’s patch token size as:

$$y_i^{\text{part}} = \arg \min_{\mathbf{c} \in \mathcal{C}} \|\mathbf{F}_{fg} - \mathbf{c}\|, \quad (5)$$

where $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_{k^*}\}$ represents the set of optimal cluster centers from k -means. We refer to these as 1^{st} order

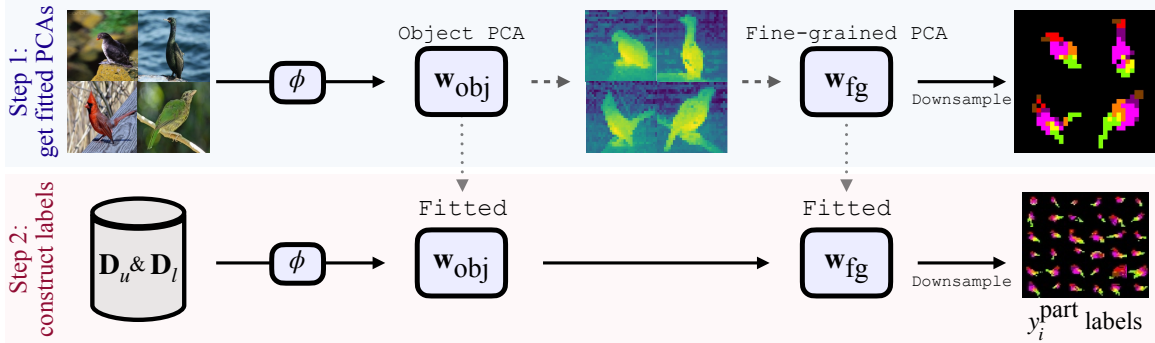


Figure 3. **Overview of part-level correspondence labels construction:** This two-step process begins by applying PCA projections to extract object and detailed features from ViT’s patch tokens using a subset of the dataset. These projections are then applied to the entire dataset to generate part-level correspondence labels.

part-level correspondence labels.

Enhancing granularity in part-level correspondence.

While 1st order part-level labels are adequate for fine-grained datasets due to the presence of shared superclasses, they may be too general for generic datasets lacking such similarities as shown in Fig. 4. To capture more intricate details in these cases, we introduce *2nd order part-level correspondence labels*. This process involves applying an additional PCA on the fine-grained features F_{fg} within each 1st order cluster. By doing so, we identify finer distinctions within each part, uncovering common features among similar but distinct parts. This 2nd order of labeling increases the granularity of part-level correspondence, enabling more precise category discovery in generic datasets where parts exhibit greater diversity and require finer resolution to discern subtle differences. We provide more discussion and analysis on our design choice of part-level correspondence labels in supp. material, Sec. S2.4.

3.2 Integrating PartCo framework with GCD method

After obtaining part-level correspondence labels, as explained in Section 3.1, we incorporate the PartCo framework, illustrated in Fig. 5 (a), into existing GCD methods. This integration is achieved through the introduction of a part-level correspondence loss $\mathcal{L}_{pc}$, which supervises the aggregated part features derived from patch tokens, thereby fostering robust part-level relationships within the ViT’s feature representations.

Guiding patch token features. For an input image x_i , we first extract its patch token features using the foundation model ϕ , yielding $F_i = \phi(x_i) \in \mathbb{R}^{N \times d}$. Utilizing the corresponding part-level correspondence labels y_i^{part} , we organize the patch features based on their assigned part categories. Specifically, for each part category $c \in \mathcal{C}$ (where $\mathcal{C}$ represents the set of all part-level categories), we aggregate the features of patches labeled as c by computing their average: $f_c = \frac{1}{|\mathcal{P}_c|} \sum_{j \in \mathcal{P}_c} F_{i,j}$, where $\mathcal{P}_c = \{j \mid y_i^{part}(j) = c\}$ denotes the set of patch indices corresponding to part category

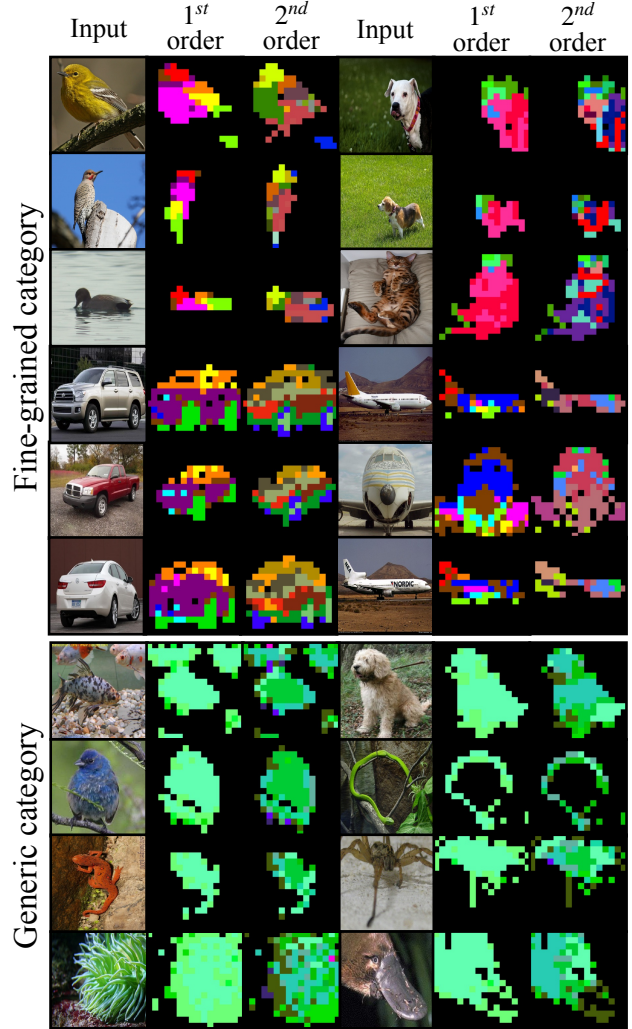


Figure 4. **Visualization of our part-level correspondence labels.** For each image, we generate both first- and second-order labels. First-order labels suffice for fine-grained datasets, while second-order labels capture additional detail for generic datasets. In practice, selecting between 1st- and 2nd-order is straightforward: datasets with subtle, intra-class differences indicate fine-grained samples, whereas datasets with pronounced, inter-class differences indicate generic samples.

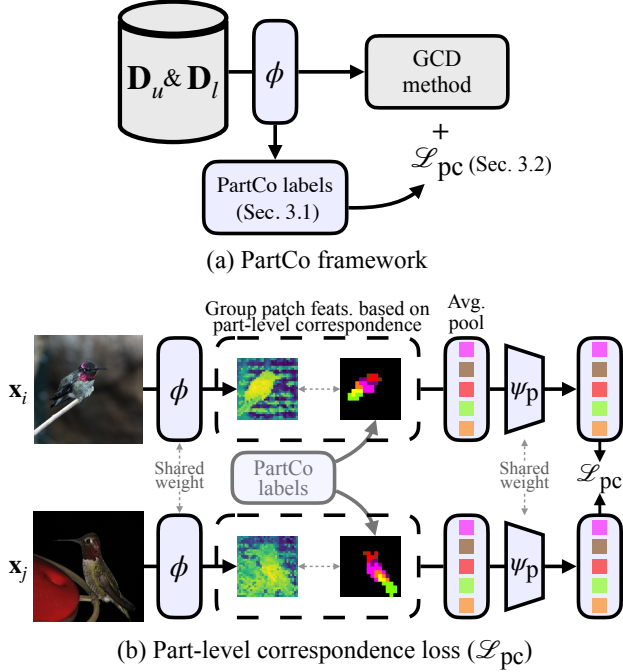


Figure 5. (a) **PartCo framework**: Introduces part-level correspondence labels as a plug-and-play module to enhance GCD methods. (b) **Part-level correspondence loss**: Depicts how part-level correspondence loss is integrated into the model to learn relationships between parts in ViT’s patch token features.

c . This pooling operation results in a set of aggregated part-level features $\{\mathbf{f}_c\}_{c \in \mathcal{C}}$, each encapsulating the information of a specific part within the image. To further refine these aggregated features, we employ a part projection head ψ_p , which projects each aggregated feature $\mathbf{f}_c$ into a new feature space: $\mathbf{h}_c = \psi_p(\mathbf{f}_c)$, where $\mathbf{h}_c \in \mathbb{R}^{d'}$ represents the projected feature for part category c , and d' is the dimensionality of the projected feature space. The projection head ψ_p is typically implemented as a multi-layer perceptron (MLP) that maps the aggregated features to a space optimized for contrastive learning. The overview of the process is shown in Fig. 5 (b).

Part-level correspondence loss. To effectively leverage these projected part-level features, we introduce a supervised part contrastive loss $\mathcal{L}_{pc}^{\text{sup}}$ that operates on the labeled data $\mathbf{D}_l$. This loss encourages features of the same part type and class to be close while separating different parts and/or classes, thereby enhancing the discriminative capability of the model at the part level. Formally, for a batch of B_l labeled samples, the supervised contrastive loss is defined:

$$\mathcal{L}_{pc}^{\text{sup}} = \frac{1}{|B_l|} \sum_{i \in B_l} \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathbb{N}_i^c|} \sum_{q \in \mathbb{N}_i^c} -\log \frac{\exp(\mathbf{h}_c \cdot \mathbf{h}_q / \tau_r)}{\sum_{j \notin \mathbb{N}_i^c} \exp(\mathbf{h}_c \cdot \mathbf{h}_j / \tau_r)}, \quad (6)$$

where $\mathbb{N}_i^c$ contains indices of labeled samples sharing the

same label y_i^l and part category c as $\mathbf{x}_i$. This loss function ensures that projected features $\mathbf{h}_c$ of the same part type and category are drawn closer in the feature space, while those of different part type and categories are repelled, thereby fostering more discriminative part-specific representations.

For parametric baselines that incorporate pseudo-labels $\mathbf{p}_i$, in Eq. 3, for unlabeled data $\mathbf{D}_u$, we extend the part-level correspondence loss to include an unsupervised part contrastive loss $\mathcal{L}_{pc}^{\text{unsup}}$. This loss, as shown in Eq. 7, operates similarly to its supervised counterpart but utilizes pseudo-labels $\mathbf{p}_i$.

$$\mathcal{L}_{pc}^{\text{unsup}} = \frac{1}{|B_u|} \sum_{i \in B_u} \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathbb{M}_i^c|} \sum_{q \in \mathbb{M}_i^c} -\log \frac{\exp(\mathbf{h}_c \cdot \mathbf{h}_q / \tau_r)}{\sum_{j \notin \mathbb{M}_i^c} \exp(\mathbf{h}_c \cdot \mathbf{h}_j / \tau_r)}, \quad (7)$$

where $\mathbb{M}_i^c$ contains indices of unlabeled samples sharing the same pseudo label and part category as $\mathbf{x}_i$. This unsupervised loss complements the supervised loss, enabling the model to learn from both labeled and unlabeled data.

Overall training objective. The integration of the PartCo framework with existing GCD methods is formalized by combining the GCD’s baseline loss in Eq. 4 with the newly introduced part-level correspondence loss. The final training objective is given by:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gcd}} + \mathcal{L}_{pc}, \quad (8)$$

where $\mathcal{L}_{pc} = (1 - \lambda_b) \mathcal{L}_{pc}^{\text{unsup}} + \lambda_b \mathcal{L}_{pc}^{\text{sup}}$ for parametric baselines, or $\mathcal{L}_{pc} = \lambda_b \mathcal{L}_{pc}^{\text{sup}}$ for non-parametric ones. By incorporating $\mathcal{L}_{pc}$, the model utilizes both global features from the [CLS] token and detailed part-specific features from the patch tokens. This dual supervision enhances the model’s ability to discover and distinguish categories with finer details.

4 Experiments

In this section, we describe our experimental setups in Sec. 4.1. Next, we present our main results in Sec. 4.2. Finally, in Sec. 4.3 we analyze the effectiveness of PartCo’s components and design choices.

4.1 Experimental setup

Datasets. We evaluate our method using several benchmark datasets. Specifically, we use the Semantic Shift Benchmark (SSB) (Vaze et al., 2022b), which includes fine-grained datasets: Caltech-UCSD Birds-200-2011 (CUB) (Wah et al., 2011), Stanford Cars (Krause et al., 2013), and FGVC-Aircraft (Maji et al., 2013). Additionally, we employ generic benchmark datasets: CIFAR10 (Krizhevsky & Hinton, 2009), CIFAR100 (Krizhevsky & Hinton, 2009), and

ImageNet-100 (Deng et al., 2009). For each dataset, we utilize the data partitioning strategy specified in Vaze et al. (2022a). This approach involves selecting a subset of all classes as the known (‘Old’) classes, denoted by $\mathbf{Y}_l$. Subsequently, 50% of the images from these known classes are allocated to the labeled dataset $\mathbf{D}_l$, and the remaining images are designated as the unlabeled dataset $\mathbf{D}_u$. Statistics of the datasets are provided in Tab. A, supp. material.

Evaluation metrics. We assess the performance of our approach using clustering accuracy (ACC; Hungarian-matched), as defined in the existing literature (Vaze et al., 2022a). The ACC for the unlabeled dataset $\mathbf{D}_u$ is calculated based on the ground-truth labels y_i^u and the predicted labels $\hat{y}_i^u$ using the following equation: $ACC = \frac{1}{|\mathbf{D}_u|} \sum_{i=1}^{|\mathbf{D}_u|} \mathbb{1}(y_i^u = h(\hat{y}_i^u))$, where h represents the optimal permutation that aligns the predicted cluster assignments with the true labels. Additionally, we report the ACC values separately for the ‘All’, ‘Old’, and ‘New’ classes.

Implementation details. We integrate our PartCo framework with the widely used parametric models SimGCD (Wen et al., 2023), SPTNet (Wang et al., 2024), FlipClass (Lin et al., 2024) and the non-parametric GCD method SelEx (Rastegar et al., 2024), employing DINO-variants (Oquab et al., 2024; Siméoni et al., 2025) pretrained weights. For SimGCD (Wen et al., 2023), the feature dimension from the backbone ϕ is set to 768. The projection head ψ , the part projection head ψ_p and the final block of ϕ are optimized using the SGD optimizer with an initial learning rate of 0.1, which decays to 0.001 following a cosine annealing schedule, and the balancing factor λ_b is fixed at 0.35. Both models are trained for 200 epochs with a batch size of 128. All input images are resized to 224×224 and augmented to match the DINO pretrained model settings. All results are on a single NVIDIA RTX 4090; PartCo adds no inference cost. The part-level label construction takes around 5–180 min depending on the dataset size (Tab. A, supp. material). Computational overhead analysis is provided in supp. material, Sec. S1.

Comparison with other methods. We compare our method with other representative GCD methods: 1) GCD (Vaze et al., 2022a); 2) SimGCD (Wen et al., 2023); 3) μ GCD (Vaze et al., 2023); 4) AMEND (Banerjee et al., 2024); 5) CiPR (Hao et al., 2024); 6) SPTNet (Wang et al., 2024); 7) ProtoGCD (Ma et al., 2025); 8) FlipClass (Lin et al., 2024); 9) SelEx (Rastegar et al., 2024); 10) APL (Dai et al., 2025); 11) AFGCD (Xu et al., 2025); 12) DebGCD (Liu & Han, 2025); 13) PartGCD (Wang et al., 2025a); 14) MOS (Peng et al., 2025); 15) SEAL (He et al., 2025b); 16) RLCD (Liu et al., 2025a); 17) AllGCD (Cao et al., 2025); 18) HypCD (Liu et al., 2025b); 19) ConGCD (Tang et al., 2025); and 20) NCGCD (Han et al., 2025). We also report k -means clustering results on frozen

DINO variants (Oquab et al., 2024; Siméoni et al., 2025).

4.2 Experimental results

Benchmark results. Tab. 1 and 2 report per-dataset results on SSB (fine-grained) and on generic datasets. PartCo delivers consistent improvements across all evaluated GCD baselines (see Fig. 6). On SSB, measured by overall ‘All’ ACC with DINOv2, PartCo-SimGCD improves by +9.8%, PartCo-SPTNet by +8.2%, PartCo-FlipClass by +4.8%, and PartCo-SelEx by +2.4%. Notably, PartCo-SelEx outperforms most existing GCD methods on both SSB and generic benchmarks across datasets, with performance on par with Hyp-SelEx but following a different design principle: HypCD improves discovery via a hyperbolic embedding transformation, whereas PartCo strengthens discovery by explicitly injecting part-level cues. This indicates that, beyond geometry-aware embeddings, explicit part structure alone can reliably boost GCD performance across datasets. The same trend persists with DINOv3 (Tabs. 1 and 2) and with CLIP features (Sec. S2.2), indicating that PartCo improves the corresponding baselines regardless of the backbone. Looking across datasets and backbones, PartCo-SimGCD improves overall ‘All’ ACC by +9.8% (DINOv2) and +3.8% (DINOv3) on SSB, while PartCo-SelEx improves by +2.4% and +2.9%, respectively (Fig. 6). On generic datasets, the gains are +0.3%/+0.9% for PartCo-SimGCD and +2.2%/+1.0% for PartCo-SelEx (v2/v3). At the per-dataset level on SSB, we observe large absolute gains across all three datasets: FGVC-Aircraft (DINOv2: +12.6%, DINOv3: +3.9%), CUB (DINOv2: +9.6%, DINOv3: +2.9%), and Stanford-Cars (DINOv2: +7.4%, DINOv3: +4.6%). On generic datasets, improvements are smaller but steady. Overall, these results show that PartCo consistently boosts diverse GCD methods and achieves strong performance across datasets and backbones.

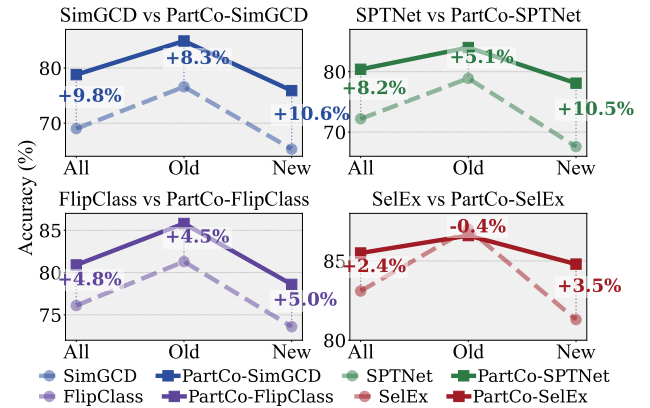


Figure 6. Average absolute % gain of PartCo over each baseline model on SSB benchmark.

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Table 1. Comparison of GCD methods on the SSB benchmark. Results are reported in ACC across the ‘All’, ‘Old’ and ‘New’ categories.

Method	Venue	Backbone	CUB			Stanford-Cars			FGVC-Aircraft			Average		
			All	Old	New	All	Old	New	All	Old	New	All	Old	New
<i>k</i> -means	-	DINOv2	67.6	60.6	71.1	29.4	24.5	31.8	18.9	16.9	19.9	38.6	34.0	40.0
GCD	CVPR’22	DINOv2	71.9	71.2	72.3	65.7	67.8	64.7	55.4	47.9	59.2	64.3	62.3	65.4
μ GCD	NeurIPS’23	DINOv2	74.0	75.9	73.1	76.1	91.0	68.9	66.3	68.7	65.1	72.1	78.6	69.0
CiPR	TMLR	DINOv2	78.3	73.4	80.8	66.7	77.0	61.8	-	-	-	-	-	-
ProtoGCD	TPAMI	DINOv2	75.7	81.5	72.9	77.6	90.5	71.5	71.1	76.3	68.5	74.8	82.7	71.0
APL	CVPR’25	DINOv2	75.1	79.1	73.2	73.4	87.6	66.7	68.8	74.1	66.6	72.4	80.3	68.8
AFGCD	ICCV’25	DINOv2	76.5	77.3	76.0	75.5	86.2	70.3	68.1	75.9	64.1	73.4	79.8	70.1
DebGCD	ICLR’25	DINOv2	77.5	80.8	75.8	75.4	87.7	69.5	71.9	76.0	69.8	74.9	81.5	71.7
MOS	CVPR’25	DINOv2	81.1	82.1	80.6	75.5	89.6	68.7	69.7	78.1	65.5	75.4	83.3	71.6
PartGCD	TMM	DINOv2	77.6	80.6	76.1	78.2	88.7	73.1	71.1	75.6	68.8	75.6	81.6	72.7
SEAL	NeurIPS’25	DINOv2	76.7	78.3	75.9	77.7	88.7	72.4	74.6	73.2	75.3	76.3	80.1	74.5
RLCD	ICML’25	DINOv2	78.7	79.5	78.3	79.5	91.8	73.5	72.6	77.3	70.3	76.9	82.9	74.0
AllGCD	ICCV’25	DINOv2	78.4	82.8	76.2	76.2	88.3	70.4	-	-	-	-	-	-
Hyp-SimGCD	CVPR’25	DINOv2	77.6	77.9	77.4	82.5	85.8	81.0	76.4	70.3	79.4	78.8	78.0	79.3
ConGCD-SelEx	ICCV’25	DINOv2	86.3	87.4	85.8	79.8	93.1	73.3	81.7	83.3	81.0	82.6	87.9	80.0
NCGCD-SelEx	NeurIPS’25	DINOv2	87.9	85.3	89.2	81.1	90.9	79.3	83.1	88.3	82.5	84.0	88.2	83.7
Hyp-SelEx	CVPR’25	DINOv2	90.7	85.3	93.4	83.8	93.3	79.2	83.4	82.0	84.1	86.0	86.9	85.5
SimGCD	ICCV’23	DINOv2	71.5	78.1	68.3	71.5	81.9	66.6	63.9	69.9	60.9	69.0	76.6	65.3
PartCo-SimGCD	Ours	DINOv2	81.1	82.4	80.5	78.9	91.5	72.8	76.5	80.9	74.4	78.8	84.9	75.9
SPTNet	ICLR’24	DINOv2	76.3	79.5	74.6	72.3	82.0	67.5	68.0	75.2	60.9	72.2	78.9	67.6
PartCo-SPTNet	Ours	DINOv2	82.6	82.3	81.8	80.1	92.0	73.5	78.6	77.6	79.0	80.4	84.0	78.1
FlipClass	NeurIPS’24	DINOv2	79.3	80.7	78.5	78.0	88.0	73.2	71.1	75.1	69.1	76.1	81.3	73.6
PartCo-FlipClass	Ours	DINOv2	85.2	86.3	84.7	80.5	92.7	74.6	77.1	78.5	76.4	80.9	85.8	78.6
SelEx	ECCV’24	DINOv2	87.4	85.1	88.5	82.2	93.7	76.7	79.8	82.3	78.6	83.1	87.0	81.3
PartCo-SelEx	Ours	DINOv2	90.6	84.5	93.2	82.5	91.8	78.0	83.4	83.6	83.3	85.5	86.6	84.8
<i>k</i> -means	-	DINOv3	69.8	64.7	72.3	59.0	50.8	63.1	40.3	35.3	42.8	56.4	50.3	59.4
SimGCD	ICCV’23	DINOv3	75.9	83.8	72.0	73.9	79.4	71.3	71.4	84.4	65.0	73.7	82.5	69.4
PartCo-SimGCD	Ours	DINOv3	78.8	83.4	76.6	78.5	86.5	74.6	75.3	81.6	72.2	77.5	83.8	74.5
SelEx	ECCV’24	DINOv3	83.5	78.1	86.2	78.8	90.3	73.3	73.9	77.3	72.4	78.7	81.9	77.3
PartCo-SelEx	Ours	DINOv3	86.1	84.4	87.0	81.7	92.1	76.6	76.9	82.2	74.2	81.6	86.2	79.3

Table 2. Comparison of GCD methods on the generic benchmark. Results are reported in ACC across the ‘All’, ‘Old’ and ‘New’ categories.

Method	Venue	Backbone	CIFAR10			CIFAR100			ImageNet-100			Average		
			All	Old	New	All	Old	New	All	Old	New	All	Old	New
<i>k</i> -means	-	DINOv2	94.9	95.2	94.8	70.9	70.8	72.1	78.3	80.5	77.2	81.4	82.2	81.4
GCD	CVPR’22	DINOv2	97.8	99.0	97.1	79.6	84.5	69.9	78.5	89.5	73.0	85.3	91.0	80.0
AMEND	WACV’24	DINOv2	97.7	96.6	98.3	83.5	83.0	84.5	87.3	95.1	83.4	89.5	91.6	88.7
CiPR	TMLR	DINOv2	99.0	98.7	99.2	90.3	89.0	93.1	88.2	87.6	88.5	92.5	91.8	93.6
SPTNet	ICLR’24	DINOv2	98.9	99.1	98.8	89.0	91.5	79.2	90.1	96.1	87.1	92.7	95.6	88.4
FlipClass	NeurIPS’24	DINOv2	99.0	98.2	99.4	91.7	90.4	94.2	91.0	96.3	88.3	93.9	95.0	94.0
DebGCD	ICLR’25	DINOv2	98.9	97.5	99.6	90.1	90.9	88.6	93.2	97.0	91.2	94.1	95.1	93.1
SEAL	NeurIPS’25	DINOv2	98.9	98.1	99.3	89.8	90.4	89.5	91.3	93.3	90.3	93.3	93.9	93.0
RLCD	ICML’25	DINOv2	99.0	98.9	99.1	91.2	91.2	91.2	92.1	96.2	90.0	94.1	95.4	93.4
Hyp-SimGCD	CVPR’25	DINOv2	98.9	97.7	99.5	91.5	90.0	94.6	91.9	96.2	89.8	94.1	94.6	94.6
Hyp-SelEx	CVPR’25	DINOv2	98.6	98.1	98.9	88.6	91.5	82.8	92.3	96.4	90.2	93.2	95.3	90.6
SimGCD	ICCV’23	DINOv2	98.7	96.7	99.7	88.5	89.2	87.2	89.9	95.5	87.1	92.4	93.8	91.3
PartCo-SimGCD	Ours	DINOv2	99.0	98.7	99.2	89.0	92.0	83.0	90.1	92.0	89.2	92.7	94.2	90.4
SelEx	ECCV’24	DINOv2	98.5	98.8	98.5	87.7	90.8	81.5	90.9	96.2	88.3	92.4	95.3	89.4
PartCo-SelEx	Ours	DINOv2	99.2	99.4	98.9	90.0	92.8	84.3	94.5	97.8	92.8	94.6	96.7	92.0
<i>k</i> -means	-	DINOv3	94.1	95.1	93.6	65.5	66.4	63.5	78.3	78.3	78.3	79.3	79.9	78.5
SimGCD	ICCV’23	DINOv3	98.4	98.7	98.2	84.3	88.3	74.3	92.2	96.5	90.0	91.6	94.8	87.5
PartCo-SimGCD	Ours	DINOv3	98.9	98.2	99.3	85.0	88.9	77.1	93.7	96.5	92.3	92.5	94.5	89.6
SelEx	ECCV’24	DINOv3	98.2	99.0	97.7	87.7	88.7	85.7	93.4	96.9	91.6	93.1	94.9	91.6
PartCo-SelEx	Ours	DINOv3	98.3	99.1	97.9	90.2	91.7	87.0	93.8	96.6	92.4	94.1	95.8	92.4

4.3 Model component analysis

Effectiveness of 1st vs. 2nd order part-level correspondence labels. We conduct an ablation study within our PartCo framework to compare 1st, and 2nd order labels, their combination, and a baseline across three fine-grained datasets and one generic dataset (Fig. 7). The results show that 1st order labels consistently achieve the highest accuracy on fine-grained datasets due to their inherently rich part-level information, which remains effective after down-sampling. In contrast, 2nd order labels, though detailed, suffer reduced performance on fine-grained datasets because downsampling limits the number of patches per label. However, on generic datasets like ImageNet-100, 2nd order labels outperform 1st order labels by providing more fine-grained information per sample, where 1st order labels lack sufficient detail. These findings highlight the importance of selecting the appropriate order level based on dataset characteristics, demonstrating PartCo’s flexibility in various scenarios.

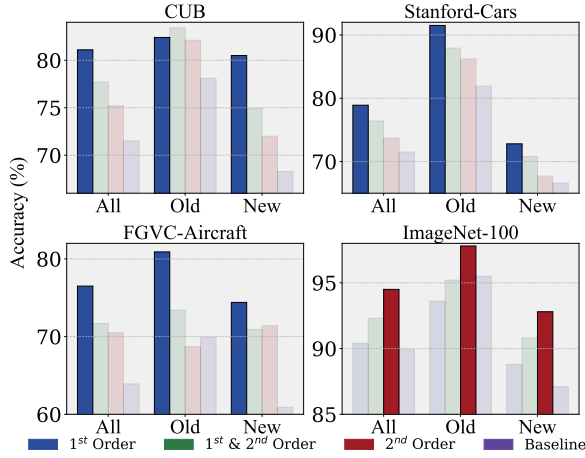


Figure 7. Ablation study investigating the impact of different order levels in part-level correspondence labels. Highest All ACC is emphasized, while lower are shown with reduced opacity.

Impact of output dimensions on part-level projection.

We ablate the output dimension d' of the part-level projection ψ_p in PartCo (Tab. 3). Across CUB and Stanford-Cars, $d' = 128$ consistently performs best, yielding the highest overall-category accuracy.

Table 3. Ablation study on part-level projection output dimensions.

Dim.	CUB			Stanford-Cars		
	All	Old	New	All	Old	New
64	79.3	76.7	80.6	76.9	88.3	71.4
128	81.1	82.4	80.5	78.9	91.5	72.8
256	79.8	80.7	79.4	76.4	90.5	69.7
512	78.2	83.1	75.7	75.6	87.5	69.9

Effect of unsupervised part-level correspondence loss. In addition to the supervised part-level correspondence loss, $\mathcal{L}_{pc}^{sup}$, we conduct an ablation study on PartCo-SimGCD to evaluate the effectiveness of the unsupervised part-level

correspondence loss, $\mathcal{L}_{pc}^{unsup}$. As demonstrated in Fig. 8, incorporating $\mathcal{L}_{pc}^{unsup}$ leads to significant improvements in category performance across datasets, highlighting the versatility and robustness of our additional loss.

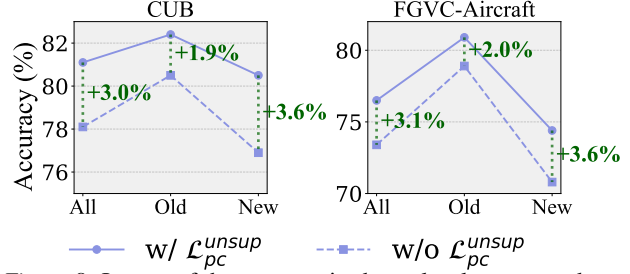


Figure 8. Impact of the unsupervised part-level correspondence loss ($\mathcal{L}_{pc}^{unsup}$).

Impact of balancing factor λ_b . As detailed in Sec. 3, to effectively leverage our part-level correspondence labels, we design a part-level correspondence loss that learns part-level relationships within the ViT’s patch features. A balancing factor λ_b is introduced to regulate the balance between our proposed loss and the baseline’s losses. In search of the optimal balancing factor, we investigate a range of weight values: 0, 0.1, 0.35, 0.7, and 1.0. The results, as shown in Tab. 4, demonstrate that a fixed weight of 0.35 is robust and consistently achieves the highest accuracy gains, highlighting PartCo’s strong practical generalization.

Table 4. Ablation study on different balancing factor λ_b values.

λ_b	CUB			Stanford-Cars		
	All	Old	New	All	Old	New
0	71.5	78.1	68.3	71.5	81.9	66.6
0.1	76.4	80.9	74.1	75.4	85.7	70.5
0.35	81.1	82.4	80.5	78.9	91.5	72.8
0.7	79.0	82.3	77.4	77.1	88.4	71.7
1.0	77.1	82.9	74.1	75.2	86.3	69.9

5 Related Work

Semi-Supervised Learning (SSL). SSL aims at learning a classifier using both labeled and unlabeled data (Chapelle et al., 2006; Zhu, 2005; Oliver et al., 2018). Most works in this domain assume that the unlabeled data contains instances from the *same* categories in the labeled data (Oliver et al., 2018). Pseudo-labeling (Rizve et al., 2021), consistency regularization (Laine & Aila, 2017; Tarvainen & Valpola, 2017; Berthelot et al., 2019; Sohn et al., 2020), and non-parametric classification (Assran et al., 2021) are among the popular methods for SSL. Most recent works further remove the assumption on the categories in the unlabeled and labeled set, (Saito et al., 2021; Huang et al., 2021; Yu et al., 2020), yet their focus is still on the performance in the labeled set.

Feature descriptors with vision foundation models. Re-

cent ViT models, particularly the DINO variants (Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025), have advanced the generation of semantically meaningful and spatially coherent features for dense visual descriptors (Amir et al., 2022). Building on DINOv1 (Caron et al., 2021), DINOv2 (Oquab et al., 2024) scales up the model size and incorporates a larger curated dataset, enhancing generalization in correspondence tasks such as part-level segmentation and zero-shot semantic correspondence (Zhang et al., 2023).

Part-level learning and semantic correspondence. Supervised fine-grained pipelines use annotated parts to normalize pose and reduce intra-class variance (Branson et al., 2014), and parts can be transferred between categories with reduced labels (Novotny et al., 2016). To reduce annotation cost, weakly / self-supervised methods learn dense features sensitive to geometry or geometrically stable for semantic matching (Novotny et al., 2017; 2018), while unsupervised approaches induce parts through shape-appearance disentangling or contrastive reconstruction (Lorenz et al., 2019; Choudhury et al., 2021). Cross-category canonicalization learns shared dense geometry without manual inter-category links (Neverova et al., 2021), and open-vocabulary part segmentation scales part reasoning using vision-language supervision and DINO-style descriptors (Sun et al., 2023; Choi et al., 2025). Unlike these lines that focus on part alignment or segmentation, we study *Generalized Category Discovery* (GCD), where a small labeled subset (known classes) coexists with unlabeled data containing both known and novel classes. Prior part-based methods typically assume part/keypoint labels (Novotny et al., 2016) or optimize matching-specific objectives (Novotny et al., 2017); while AnchorNet transfers to unseen classes for matching (Novotny et al., 2017), these works do not discover or cluster unknown categories without labels. In contrast, *PartCo* distills *part-level observations* into GCD via a tailored correspondence loss, yielding robust, part-aware features under mixed supervision and enabling semantically coherent clusters for unseen categories without part annotations or open-vocabulary text labels, while benefiting from the explicit inductive biases revealed by part-level correspondences.

Category Discovery. Novel Class Discovery (NCD) (Han et al., 2019) facilitates knowledge transfer from known to unseen categories, by considering it as a transfer clustering problem. Since its introduction, various methods are developed to advance NCD (Han et al., 2020; 2021; Jia et al., 2021; Zhao & Han, 2021; Zhong et al., 2021; Fini et al., 2021). Generalized Category Discovery (GCD) (Vaze et al., 2022a) extends NCD by incorporating unlabeled data from both known and unknown classes, presenting additional challenges. Subsequent research on GCD proposes diverse strategies to address these complexities (Cao et al., 2022; Joseph et al., 2022; Pu et al., 2023; Hao et al., 2024; Cendra et al., 2024; Wang et al., 2025c; Liu & Han, 2025;

Han et al., 2025). For example, SimGCD (Wen et al., 2023) introduces a parametric classifier with mean entropy regularization, while GPC (Zhao et al., 2023) utilizes Gaussian mixture models to learn robust representations and estimate the number of unknown categories. SPTNet (Wang et al., 2024) employs spatial prompt tuning to enhance focus on specific object parts, improving knowledge transfer in GCD tasks. Recently, FlipClass (Lin et al., 2024) dynamically updates the teacher model to align with the student’s attention, ensuring consistency in all classes, SelEx (Rastegar et al., 2024) achieves strong performance in fine-grained datasets through hierarchical semi-supervised k -means clustering, NC-GCD (Han et al., 2025) leverages a neural collapse-inspired framework for generalized category discovery, and HypCD (Liu et al., 2025b) achieves strong performances by learning representation in hyperbolic space. Additionally, category discovery is explored in various contexts, including multi-modal settings (Jia et al., 2021), continual learning (Zhang et al., 2022; Cendra et al., 2024), federated environments (Pu et al., 2024), and handling domain shifts (Wang et al., 2025c). More recently, APL (Dai et al., 2025) and PartGCD (Wang et al., 2025a) have explored incorporating part-level information into GCD. APL learns part queries guided by DINO priors and replaces the backbone’s classification features with an aggregated part representation, while PartGCD introduces adaptive part decomposition based on Gaussian Mixture Models together with a part discrepancy regularization that explicitly separates part features. In contrast, PartCo introduces explicit part-level correspondence priors derived from frozen pretrained features and uses them as a training-only supervisory signal, without modifying the original architecture or inference path of the host baseline. This distinction makes PartCo a novel yet simple and flexible framework that can be integrated into different GCD methods while improving category discovery through explicit part-level correspondence priors.

6 Conclusion

In this paper, we introduce PartCo, a learning framework for GCD that integrates explicit part-level visual feature correspondences. Unlike traditional GCD methods that rely solely on semantic labels, PartCo leverages the detailed composition of object features to improve category understanding and discovery. Our experiments on multiple benchmark datasets demonstrate that our framework significantly boosts existing GCD methods. Its seamless integration with current approaches without major modifications highlights its practicality and broad applicability. By focusing on part-level relationships, PartCo not only increases discovery accuracy but also provides deeper insights into the visual structures underlying semantic labels. Overall, PartCo bridges the gap between semantic labels and part-level feature compositions, outperforming most existing GCD methods.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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References

- Amir, S., Gandelsman, Y., Bagon, S., and Dekel, T. Deep vit features as dense visual descriptors. In *ECCV workshop*, 2022. 9
- Assran, M., Caron, M., Misra, I., Bojanowski, P., Joulin, A., Ballas, N., and Rabbat, M. Semi-supervised learning of visual features by non-parametrically predicting view assignments with support samples. In *ICCV*, 2021. 8
- Banerjee, A., Kallooriyakath, L. S., and Biswas, S. Amend: Adaptive margin and expanded neighborhood for efficient generalized category discovery. In *WACV*, 2024. 6
- Berthelot, D., Carlini, N., Goodfellow, I., Papernot, N., Oliver, A., and Raffel, C. Mixmatch: A holistic approach to semi-supervised learning. In *NeurIPS*, 2019. 8
- Branson, S., Van Horn, G., Belongie, S., and Perona, P. Bird species categorization using pose normalized deep convolutional nets. In *BMVC*, 2014. 9
- Cao, K., Brbic, M., and Leskovec, J. Open-world semi-supervised learning. In *ICLR*, 2022. 9
- Cao, X., Chen, K., Yang, F., Zheng, X., Tian, Y., and Lu, Y. Allgcd: Leveraging all unlabeled data for generalized category discovery. In *ICCV*, 2025. 6
- Caron, M., Touvron, H., Misra, I., Jegou, H., Mairal, J., Bojanowski, P., and Joulin, A. Emerging properties in self-supervised vision transformers. In *ICCV*, 2021. 2, 9, 20
- Cendra, F. J., Zhao, B., and Han, K. Promptccd: Learning gaussian mixture prompt pool for continual category discovery. In *ECCV*, 2024. 9
- Chapelle, O., Scholkopf, B., and Zien, A. Semi-supervised learning. *MIT Press*, 2006. 8
- Choi, J., Lee, S., Lee, M., Lee, S., and Shim, H. Fine-grained image-text correspondence with cost aggregation for open-vocabulary part segmentation. In *CVPR*, 2025. 9
- Choudhury, S., Laina, I., Rupprecht, C., and Vedaldi, A. Un-supervised part discovery from contrastive reconstruction. In *NeurIPS*, 2021. 9
- Dai, Q., Huang, H., Wu, Y., and Yang, S. Adaptive part learning for fine-grained generalized category discovery: A plug-and-play enhancement. In *CVPR*, 2025. 6, 9
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009. 6, 14
- Fini, E., Sangineto, E., Lathuiliere, S., Zhong, Z., Nabi, M., and Ricci, E. A unified objective for novel class discovery. In *ICCV*, 2021. 9
- Fu, S., Hamilton, M., Brandt, L., Feldman, A., Zhang, Z., and Freeman, W. T. Featup: A model-agnostic framework for features at any resolution. In *ICLR*, 2024. 25
- Han, J., Wang, S., He, Y., Ding, C., Wang, Q., Gao, X., Dong, S., and Gong, Y. Consistent supervised-unsupervised alignment for generalized category discovery. In *NeurIPS*, 2025. 6, 9
- Han, K., Vedaldi, A., and Zisserman, A. Learning to discover novel visual categories via deep transfer clustering. In *ICCV*, 2019. 2, 9
- Han, K., Rebuffi, S.-A., Ehrhardt, S., Vedaldi, A., and Zisserman, A. Automatically discovering and learning new visual categories with ranking statistics. In *ICLR*, 2020. 9
- Han, K., Rebuffi, S.-A., Ehrhardt, S., Vedaldi, A., and Zisserman, A. Autonovel: Automatically discovering and learning novel visual categories. *IEEE TPAMI*, 2021. 2, 9
- Hao, S., Han, K., and Wong, K.-Y. K. Cipr: An efficient framework with cross-instance positive relations for generalized category discovery. *TMLR*, 2024. 2, 6, 9, 18
- He, Z., Liu, Y., and Han, K. Category discovery: An open-world perspective. *arXiv preprint arXiv:2509.22542*, 2025a. 1
- He, Z., Liu, Y., and Han, K. Seal: Semantic-aware hierarchical learning for generalized category discovery. In *NeurIPS*, 2025b. 6
- Huang, J., Fang, C., Chen, W., Chai, Z., Wei, X., Wei, P., Lin, L., and Li, G. Trash to treasure: harvesting ood data with cross-modal matching for open-set semi-supervised learning. In *ICCV*, 2021. 8
- Jia, X., Han, K., Zhu, Y., and Green, B. Joint representation learning and novel category discovery on single-and multi-modal data. In *ICCV*, 2021. 9

- Joseph, K., Paul, S., Aggarwal, G., Biswas, S., Rai, P., Han, K., and Balasubramanian, V. N. Novel class discovery without forgetting. In *ECCV*, 2022. 9
- Krause, J., Stark, M., Deng, J., and Fei-Fei, L. 3d object representations for fine-grained categorization. In *ICCV workshop*, 2013. 5, 14, 18, 20
- Krizhevsky, A. and Hinton, G. Learning multiple layers of features from tiny images. *Master's thesis, Department of Computer Science, University of Toronto*, 2009. 5, 14
- Laine, S. and Aila, T. Temporal ensembling for semi-supervised learning. In *ICLR*, 2017. 8
- Lin, H., An, W., Wang, J., Chen, Y., Tian, F., Wang, M., Wang, Q., Dai, G., and Wang, J. Flipped classroom: Aligning teacher attention with student in generalized category discovery. In *NeurIPS*, 2024. 6, 9
- Liu, D., Tan, Z., Zhao, L., Zhang, Z., Fang, X., and Huang, W. Generalized category discovery via reciprocal learning and class-wise distribution regularization. In *ICML*, 2025a. 6
- Liu, Y. and Han, K. Debgcd: Debaised learning with distribution guidance for generalized category discovery. In *ICLR*, 2025. 6, 9
- Liu, Y., He, Z., and Han, K. Hyperbolic category discovery. In *CVPR*, 2025b. 6, 9
- Locatello, F., Weissenborn, D., Unterthiner, T., Mahendran, A., Heigold, G., Uszkoreit, J., Dosovitskiy, A., and Kipf, T. Object-centric learning with slot attention. In *NeurIPS*, 2020. 25
- Lorenz, D., Bereska, L., Milbich, T., and Ommer, B. Un-supervised part-based disentangling of object shape and appearance. In *CVPR*, 2019. 9
- Ma, S., Zhu, F., Zhang, X.-Y., and Liu, C.-L. Protogcd: Unified and unbiased prototype learning for generalized category discovery. *IEEE TPAMI*, 2025. 6
- Maji, S., Rahtu, E., Kannala, J., Blaschko, M., and Vedaldi, A. Fine-grained visual classification of aircraft. *arXiv preprint arXiv:1306.5151*, 2013. 5, 14
- Neverova, N., Sanakoyeu, A., Labatut, P., Novotny, D., and Vedaldi, A. Discovering relationships between object categories via universal canonical maps. In *CVPR*, 2021. 9
- Novotny, D., Larlus, D., and Vedaldi, A. I have seen enough: Transferring parts across categories. In *BMVC*, 2016. 9
- Novotny, D., Larlus, D., and Vedaldi, A. AnchorNet: A weakly supervised network to learn geometry-sensitive features for semantic matching. In *CVPR*, 2017. 9
- Novotny, D., Albanie, S., Larlus, D., and Vedaldi, A. Self-supervised learning of geometrically stable features through probabilistic introspection. In *CVPR*, 2018. 9
- Oliver, A., Odena, A., Raffel, C. A., Cubuk, E. D., and Goodfellow, I. Realistic evaluation of deep semi-supervised learning algorithms. In *NeurIPS*, 2018. 8
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H. V., Szafraniec, M., Khalidov, V., Fernandez, P., HAZIZA, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., Synnaeve, G., Xu, H., Jegou, H., Mairal, J., Labatut, P., Joulin, A., and Bojanowski, P. DINOv2: Learning robust visual features without supervision. *TMLR*, 2024. 2, 3, 6, 9, 17, 18
- Ouldnooghi, R., Kuo, C.-W., and Kira, Z. Clip-gcd: Simple language guided generalized category discovery. *arXiv preprint arXiv:2305.10420*, 2023. 15
- Parkhi, O. M., Vedaldi, A., Zisserman, A., and Jawahar, C. Cats and dogs. In *CVPR*, 2012. 14, 15
- Peng, Z., Ma, J., Sun, Z., Yi, R., Song, H., Tan, X., and Ma, L. Mos: Modeling object-scene associations in generalized category discovery. In *CVPR*, 2025. 6
- Pu, N., Zhong, Z., and Sebe, N. Dynamic conceptional contrastive learning for generalized category discovery. In *CVPR*, 2023. 9
- Pu, N., Li, W., Ji, X., Qin, Y., Sebe, N., and Zhong, Z. Federated generalized category discovery. In *CVPR*, 2024. 9
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 15
- Rastegar, S., Salehi, M., Asano, Y. M., Doughty, H., and Snoek, C. G. M. Selex: Self-expertise in fine-grained generalized category discovery. In *ECCV*, 2024. 6, 9, 18, 20
- Rizve, M. N., Duarte, K., Rawat, Y. S., and Shah, M. In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning. In *ICLR*, 2021. 8
- Saito, K., Kim, D., and Saenko, K. Openmatch: Open-set consistency regularization for semi-supervised learning with outliers. In *NeurIPS*, 2021. 8
- Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025. 2, 3, 6, 9

- Sohn, K., Berthelot, D., Carlini, N., Zhang, Z., Zhang, H., Raffel, C. A., Cubuk, E. D., Kurakin, A., and Li, C.-L. Fixmatch: Simplifying semi-supervised learning with consistency and confidence. In *NeurIPS*, 2020. 8
- Sun, P., Chen, S., Zhu, C., Xiao, F., Luo, P., Xie, S., and Yan, Z. Going denser with open-vocabulary part segmentation. In *ICCV*, 2023. 9
- Tan, K. C., Liu, Y., Ambrose, B., Tulig, M., and Belongie, S. The herbarium challenge 2019 dataset. In *CVPR workshop*, 2019. 14, 15
- Tang, L., Huang, K., Chen, C., Yuan, Y., Li, C., Tu, X., Ding, X., and Huang, Y. Dissecting generalized category discovery: Multiplex consensus under self-deconstruction. In *ICCV*, 2025. 6
- Tarvainen, A. and Valpola, H. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *NeurIPS*, 2017. 8
- Vaze, S., Han, K., Vedaldi, A., and Zisserman, A. Generalized category discovery. In *CVPR*, 2022a. 1, 2, 6, 9, 14, 18
- Vaze, S., Han, K., Vedaldi, A., and Zisserman, A. The semantic shift benchmark. In *ICML workshop*, 2022b. 5
- Vaze, S., Vedaldi, A., and Zisserman, A. No representation rules them all in category discovery. In *NeurIPS*, 2023. 2, 6
- Wah, C., Branson, S., Welinder, P., Perona, P., and Belongie, S. The caltech-ucsd birds-200-2011 dataset. *California Institute of Technology*, 2011. 5, 14, 18, 20
- Wang, E., Peng, Z., Xie, Z., Lu, H., Yang, F., and Liu, X. Learning part knowledge to facilitate category understanding for fine-grained generalized category discovery. *IEEE TMM*, 2025a. 6, 9
- Wang, E., Peng, Z., Xie, Z., Yang, F., Liu, X., and Cheng, M.-M. Get: Unlocking the multi-modal potential of clip for generalized category discovery. In *CVPR*, 2025b. 15
- Wang, H., Vaze, S., and Han, K. Sptnet: An efficient alternative framework for generalized category discovery with spatial prompt tuning. In *ICLR*, 2024. 1, 2, 6, 9, 15
- Wang, H., Vaze, S., and Han, K. Hilo: A learning framework for generalized category discovery robust to domain shifts. In *ICLR*, 2025c. 9
- Wen, X., Zhao, B., and Qi, X. Parametric classification for generalized category discovery: A baseline study. In *ICCV*, 2023. 2, 6, 9, 15, 20
- Wimmer, T., Truong, P., Rakotosaona, M.-J., Oechsle, M., Tombari, F., Schiele, B., and Lenssen, J. E. Anyup: Universal feature upsampling. In *ICLR*, 2026. 25
- Xu, Q., Hu, Z., Duan, Y., Pei, E., and Tai, Y. A hidden stumbling block in generalized category discovery: Distracted attention. In *ICCV*, 2025. 6
- Yang, M., Yin, J., Gu, Y., Deng, C., Zhang, H., and Zhu, H. Consistent prompt tuning for generalized category discovery. *IJCV*, 2025. 15
- Yu, Q., Ikami, D., Irie, G., and Aizawa, K. Multi-task curriculum framework for open-set semi-supervised learning. In *ECCV*, 2020. 8
- Zhang, J., Herrmann, C., Hur, J., Polania Cabrera, L., Jampani, V., Sun, D., and Yang, M.-H. A tale of two features: Stable diffusion complements dino for zero-shot semantic correspondence. In *NeurIPS*, 2023. 9, 17
- Zhang, X., Jiang, J., Feng, Y., Wu, Z.-f., Zhao, X., Wan, H., Tang, M., Jin, R., and Gao, Y. Grow and merge: a unified framework for continuous categories discovery. In *NeurIPS*, 2022. 9
- Zhao, B. and Han, K. Novel visual category discovery with dual ranking statistics and mutual knowledge distillation. In *NeurIPS*, 2021. 9
- Zhao, B., Wen, X., and Han, K. Learning semi-supervised gaussian mixture models for generalized category discovery. In *ICCV*, 2023. 2, 9, 18
- Zheng, H., Pu, N., Li, W., Sebe, N., and Zhong, Z. Textual knowledge matters: Cross-modality co-teaching for generalized visual class discovery. In *ECCV*, 2024. 15
- Zhong, Z., Fini, E., Roy, S., Luo, Z., Ricci, E., and Sebe, N. Neighborhood contrastive learning for novel class discovery. In *CVPR*, 2021. 9
- Zhu, X. J. Semi-supervised learning literature survey. *University of Wisconsin-Madison Department of Computer Sciences*, 2005. 8

PartCo: Part-Level Correspondence Priors Enhance Category Discovery

–Supplementary Material–

Overview. We describe more experimental details (Sec. S1), covering dataset statistics, runtime measurements, and computational overhead analysis. Building on this setup, we present additional quantitative results (Sec. S2), including evaluations on more datasets, experiments using CLIP backbone, an analysis of integrating implicit and explicit part-level cues, and a deeper examination of part-level correspondence labels, comparisons of part-level label construction strategies, studies on unknown-category estimation. We complement these findings with additional qualitative results (Sec. S3), featuring extended part-label visualizations and a qualitative analysis of PartCo attention maps. To contextualize performance, we discuss success and failure cases of our framework (Sec. S4). Finally, we include a discussion of our framework’s limitations and future work (Sec. S5).

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S1 Additional Experimental Details

S1.1 Benchmark datasets

For each benchmark dataset, we follow the data splitting approach outlined in Vaze et al. (2022a). In this approach, 50% of the classes are designated as ‘Old’, except for CIFAR100, which selects 80% of the classes. Subsequently, 50% of the images from the known classes form the labeled dataset $\mathbf{D}_l$, while the remaining images are assigned to the unlabeled dataset $\mathbf{D}_u$. The statistics of all datasets used in this study are presented in Tab. A.

S1.2 Time analysis for part-level correspondence labels construction

In Sec. 3.1 of the main paper, we describe the methodology for generating part-level correspondence labels. The overall time required to construct these labels for each dataset is detailed in the last column of Tab. A. As illustrated, our label construction process is efficient, with execution times ranging from approximately 5 to 180 minutes depending on the dataset’s size and complexity. Importantly, this time cost is negligible compared to typical model training durations, and it is incurred solely during the training phase. Consequently, the label construction does not introduce any additional latency during the inference phase. While it is feasible to integrate the label construction directly into the training pipeline, we have opted to separate these steps to simplify the implementation and facilitate easier experimentation.

Table A. Dataset statistics. We specify the number of classes in the labeled and unlabeled sets as $M = |\mathbf{Y}_l|$ and $K = |\mathbf{Y}_l \cup \mathbf{Y}_u|$, respectively, along with the image counts $|\mathbf{D}_l|$ and $|\mathbf{D}_u|$. In the last column, we also provide the time taken (minutes) to construct PartCo’s labels.

Dataset	$ \mathbf{D}_l $	M	$ \mathbf{D}_u $	K	PartCo label time (min)
CUB (Wah et al., 2011)	1.5K	100	4.5K	200	6 m
Stanford-Cars (Krause et al., 2013)	2.0K	98	6.1K	196	5 m
FGVC-Aircraft (Maji et al., 2013)	1.7K	50	5.0K	100	7 m
CIFAR10 (Krizhevsky & Hinton, 2009)	12.5K	5	37.5K	10	30 m
CIFAR100 (Krizhevsky & Hinton, 2009)	20.0K	80	30.0K	100	30 m
ImageNet-100 (Deng et al., 2009)	31.9K	50	95.3K	100	180 m
Oxford-Pet (Parkhi et al., 2012)	0.9K	19	2.7K	37	5 m
Herbarium19 (Tan et al., 2019)	8.9K	341	25.4K	683	118 m

S1.3 Computational overhead analysis

We report concrete training and compute statistics in Tab. B. Compared to SimGCD, PartCo-SimGCD incurs only modest overhead on CUB / Stanford Cars / CIFAR100: training time increases by just 5%, 2.5%, and 4.2% respectively, peak memory (with 128 batch size) rises from 8.6 GB to 12.0 GB, and GFLOPs grow by only 15%. This extra cost mainly comes from computing patch-level correspondences and the additional patch projection head. Importantly, inference remains unchanged: at test time, PartCo uses the same backbone features and clustering pipeline as the underlying baseline methods. By introducing this small computational overhead over the baseline, PartCo significantly improves current baselines across benchmarks and pretrained backbones.

Table B. Computational overhead results. Computational overhead of PartCo-SimGCD relative to SimGCD: modest increases in training time, peak memory, and GFLOPs; inference pipeline remains unchanged.

Method	Training time			Peak training memory usage	GFLOPs
	CUB	Stanford Cars	CIFAR100		
SimGCD (baseline)	3h 22m	4h 45m	30h 7m	8.6 GB	54.1
PartCo-SimGCD (Ours)	3h 32m (+5%)	4h 52m (+2.5%)	31h 22m (+4.2%)	12.0 GB	62.6 (+15%)

S2 Additional Quantitative Results

S2.1 Experiments on additional datasets

To further assess the effectiveness of our PartCo framework, we conducted evaluations on two additional fine-grained datasets: Oxford-Pet (Parkhi et al., 2012) and Herbarium19 (Tan et al., 2019). The Oxford-Pet dataset is particularly challenging due to its diverse assortment of cat and dog species combined with limited data availability. In contrast, Herbarium19 is a botanical research dataset that includes a wide variety of plant types, characterized by its long-tailed distribution and detailed categorization. The details of these two datasets are shown in Tab. A.

As shown in Tab. C, our PartCo-enhanced models consistently outperform the baseline method across all categories. Specifically, PartCo-SimGCD achieves an impressive accuracy of 95.2% on the ‘All’ category of the Oxford-Pet dataset, significantly surpassing the SimGCD baseline’s 86.2%. Similarly, on the Herbarium19 dataset, PartCo-SimGCD attains an accuracy of 55.5%, outperforming the baseline’s 48.6%. Overall, these results demonstrate that our PartCo framework effectively enhances existing baseline models, even when applied to more challenging fine-grained and long-tailed datasets.

Table C. Enhancement of baseline GCD methods with PartCo framework. Performance on the Oxford-Pet (Parkhi et al., 2012) and Herbarium19 (Tan et al., 2019) datasets using DINOv2. Results are reported in ACC across the ‘All’, ‘Old’ and ‘New’ categories. † denotes results implemented by us.

Method	Oxford-Pet			Herbarium19		
	All	Old	New	All	Old	New
SimGCD	86.2	85.4	86.6	48.6	64.8	39.9
PartCo-SimGCD (Ours)	95.2	92.7	96.6	55.5	68.0	48.7
FlipClass†	91.4	88.0	93.2	55.8	68.1	49.2
PartCo-FlipClass (Ours)	95.3	93.7	96.1	57.4	69.0	51.2
SelEx†	91.5	96.7	88.6	43.1	54.1	37.2
PartCo-SelEx (Ours)	92.7	96.9	90.4	45.9	57.3	39.7

S2.2 Experiment on CLIP backbone

To assess PartCo’s generalization beyond DINO variants, we run additional experiments using only CLIP (Radford et al., 2021) ViT-B/16 vision encoder (see Tab. D). Even with the image encoder alone, PartCo consistently and substantially improves the SimGCD baseline across multiple datasets, yielding strong overall performance (with $\sim 5\%$ accuracy gains). Note that competing methods (Zheng et al., 2024; Wang et al., 2025b; Ouldnoughi et al., 2023; Yang et al., 2025) use both CLIP’s image and text encoders. This suggests that our framework does not hinge on a particular pretraining strategy (e.g., DINO) but instead leverages a more general property of modern vision transformers: their ability to provide reasonably structured patch-level features. As with any method that builds on a pretrained backbone, PartCo will of course inherit some biases of the underlying foundation model. However, the empirical results with both DINOv2/v3 and CLIP indicate that PartCo is robust across different pretrained backbones, and that its benefits are not confined to a single pretrained family.

Table D. Comparison of GCD methods on the SSB benchmark datasets using CLIP backbone. Results are reported in ACC across the ‘All’, ‘Old’ and ‘New’ categories.

Method	Encoder	Backbone	CUB			Stanford-Cars			FGVC-Aircraft			Average		
			All	Old	New	All	Old	New	All	Old	New	All	Old	New
GCD	Image	CLIP	57.6	65.2	53.8	65.1	75.9	59.8	45.3	44.4	45.8	56.0	61.8	53.1
TextGCD	Image & Text	CLIP	76.6	80.6	74.7	86.9	87.4	86.7	-	-	-	-	-	-
GET	Image & Text	CLIP	77.0	78.1	76.4	78.5	86.8	74.5	58.9	59.6	58.5	71.5	74.8	69.8
CLIP-GCD	Image & Text	CLIP	62.8	77.1	55.7	70.6	88.2	62.2	50.0	56.6	46.5	61.1	74.0	54.8
CPT	Image & Text	CLIP	70.1	73.5	68.4	74.2	84.3	69.3	-	-	-	-	-	-
SimGCD	Image	CLIP	71.7	76.5	69.4	70.0	83.4	63.5	54.3	58.4	52.2	65.3	72.8	61.7
PartCo-SimGCD (Ours)	Image	CLIP	73.4	80.2	70.0	76.5	89.6	70.2	61.6	62.9	60.9	70.5	77.6	67.0

S2.3 Part-level learning boosts category discovery

We assess how part-level learning improves GCD by adding (i) an implicit part learner (SPTNet (Wang et al., 2024)) and (ii) our explicit framework (PartCo) to the SimGCD baseline (Wen et al., 2023) with DINOv2. Results on CUB and Stanford-Cars datasets are summarized in Tab. E. As demonstrated, both approaches provide clear benefits. Adding

SPTNet to SimGCD improves the overall accuracy. Using PartCo alone brings larger gains (81.1% on CUB; 78.9% on Stanford-Cars). Combining SPTNet with PartCo outperforms other methods, indicating strong complementarity: implicit cues from SPTNet and explicit part constraints from PartCo enhance feature quality and category separation in distinct ways. In summary, our explicit PartCo not only outperforms the baseline and the implicit SPTNet on its own, but also further amplifies the gains of the implicit framework when combined, underscoring that explicit part learning provides complementary supervision that unlocks additional improvements in category discovery.

Table E. **Implicit vs. explicit part-level learning.** Study on implicit (SPTNet) and explicit (PartCo) part-level learning frameworks on baseline parametric model: SimGCD.

SimGCD baseline		CUB			Stanford-Cars		
+ PartCo (Ours)	+ SPTNet	All	Old	New	All	Old	New
✗	✗	71.5	78.1	68.3	71.5	81.9	66.6
✗	✓	76.3	79.5	74.6	72.5	82.0	67.5
✓	✗	<u>81.1</u>	82.4	<u>80.5</u>	<u>78.9</u>	<u>91.5</u>	<u>72.8</u>
✓	✓	82.6	<u>82.3</u>	81.8	80.1	92.0	73.5

S2.4 Further analysis on part-level correspondence labels

Comparison between PartCo’s labels and annotated part labels. To verify the effectiveness and quality of our part-level correspondence labels, we compare our constructed CUB’s part correspondence labels (before downsample) against the manually annotated parts provided by CUB dataset and report the results in Tab. F. For each visible ground-truth part, we check whether the assigned part labels match the corresponding CUB part through the Hungarian matching algorithm. The results show that our 1st and 2nd order part-level correspondence labels achieve overall accuracies of 80.8% and 77.5%, respectively, with especially strong alignment on key semantic parts such as *back*, *beak*, *belly*, *leg*, *wing*, and *tail*. Moreover, the significant performance gains reported in the main paper (Tab. 1 and Tab. 2) suggest that the part-level correspondence labels are of sufficiently high quality to provide a reliable supervisory signal for our part-level correspondence learning framework.

Table F. Comparison with CUB ground-truth part labels.

CUB’s part index	No. visible parts	CUB’s part-level correspondence labels	
		1 st part labels ACC	2 nd part labels ACC
Part 1: back	9064	96.2	97.5
Part 2: beak	11745	97.9	91.0
Part 3: belly	10347	76.2	89.2
Part 4: crown	11580	86.4	89.8
Part 5: forehead	11603	50.2	43.1
Part 6: leg (left/right)	17008	85.2	75.7
Part 7: wing (left/right)	13255	67.6	69.7
Part 8: tail	10961	89.5	70.9
<i>Overall</i>	95563	80.8	77.5

Why further expanding PartCo’s order levels hurts category discovery. In the main paper, we show that first-order part labels and second-order part labels yield the best gains on fine-grained and coarse-grained datasets respectively (see Fig. 7). In this section, we ask whether pushing to even higher orders continues to help. To examine this, we extend PartCo to 3rd order part-level correspondences by mirroring our second-order construction: we apply an additional PCA on the fine-grained features within each second-order cluster to obtain finer partitions. As visualized for CUB in Fig. A, this produces over-fragmentation: the same semantic part (e.g., a bird’s head or wing) is split across several distinct labels, yielding redundant, noisy correspondences that are less semantically meaningful and harder to transfer.

We further quantitatively examine this in Tab. G, by utilizing 3rd order labels in our framework for CUB and ImageNet100 datasets; performance consistently drops. With a fixed token budget in the transformer (e.g., 196 patch tokens), the over-clustered supervision spreads attention too thin, impeding effective part-level correspondence learning compared to our current design. Taken together, these results suggest a practical principle: 1st order and 2nd order correspondences strike the right balance, capturing part identity for both fine-grained and generic settings, whereas higher orders over-cluster and lead

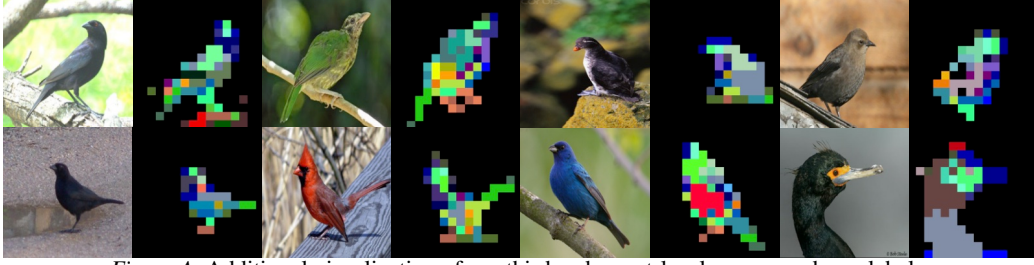


Figure A. Additional visualization of our third-order part-level correspondence labels.

to ineffective part correspondence learning.

Table G. **Ablation of part-level order granularity.** Extending PartCo to third-order labels consistently hurts accuracy across datasets; first-order and second-order labels provide the best improvement gains for fine-grained and generic datasets respectively.

Method	CUB			ImageNet100		
	All	Old	New	All	Old	New
baseline	71.5	78.1	68.3	89.9	95.5	87.1
+1 st order labels	81.1	82.4	80.5	90.4	93.6	88.8
+2 nd order labels	75.2	82.1	72.0	94.6	96.7	92.0
+3 rd order labels	72.7	81.3	68.4	91.8	94.6	90.4

Comparison of different approaches for part-level label construction. In this work, we use PCA + DINOv2 (Oquab et al., 2024) features for the following reasons: **(1) Generalization.** PCA with DINOv2 offers excellent off-the-shelf generalization for generating part-level correspondences without the need for additional tuning or adaptation. According to (Zhang et al., 2023), DINOv2 outperforms other foundation models like DINO and Stable Diffusion (SD) models, and is only slightly less effective than combining DINOv2 + SD. **(2) Efficiency.** Incorporating SD-based features, or a combination of SD and DINO-based models, significantly increases computational costs and memory usage (Zhang et al., 2023). This makes the part-label construction process inefficient. By using DINOv2 solely, our approach remains both computationally and model-efficient.

Moreover, we construct labels by augmenting DINOv2 features with SD features to compute PCA, following (Zhang et al., 2023). Because SD requires an inference denoising step, this procedure is computationally heavy: on CUB it takes around 108 minutes, whereas our PCA + DINOv2 pipeline takes around 6 minutes (Tab. A, supp. material), making SD-DINOv2 impractical for larger datasets (e.g., ImageNet-100). We further compared performance using SD-DINOv2-based labels against our original labels. As shown in Tab. H, the ‘All’ ACC differences are marginal (about 0.2–0.6 percentage points) while our PCA + DINOv2 label construction is substantially faster and more efficient.

Table H. **Influence of different part-level correspondence labels construction techniques.** Performance on the CUB and Stanford-Cars datasets with different part-level correspondence labels generated by Our method vs. SD-DINOv2 (Zhang et al., 2023). Results are reported in ACC across the ‘All’, ‘Old’ and ‘New’ categories.

PartCo-SimGCD	CUB			Stanford-Cars		
	All	Old	New	All	Old	New
w/ DINOv2 (Ours)	81.1	82.4	80.5	78.9	91.5	72.8
w/ SD-DINOv2	80.9	79.7	81.6	78.3	90.0	72.9

Evaluating background suppression vs. explicit part correspondence. To verify whether explicit part modeling is necessary, we add ablation where we simply mask background regions on both labeled and unlabeled images using our background mask obtained during our part-level correspondence label construction and train the baselines using foreground-only features, without any explicit part correspondence learning. As shown in Tab. I, for SimGCD baseline tested on CUB and Stanford Cars, the foreground-only variant yields only modest gains over the original baseline (e.g., CUB: 71.5 → 73.7 on “All”, and Stanford Cars: 71.5 → 73.0 on “All”), indicating that removing background alone provides limited benefits. In contrast, introducing explicit part-level correspondence prior via PartCo framework leads to substantially larger improvements on the same backbones and dataset (see Tab. 1 and Tab. 2), showing that the advantage does not come merely from suppressing background noise but from explicitly aligning semantically meaningful object parts across images.

Table I. Foreground-only ablation vs. PartCo explicit part correspondence learning.

Method	CUB			Stanford Cars		
	All	Old	New	All	Old	New
SimGCD (baseline)	71.5	78.1	68.3	71.5	81.9	66.6
SimGCD (foreground only)	73.7	75.4	72.8	73.0	85.6	66.9
PartCo-SimGCD (Ours)	81.1	82.4	80.5	78.9	91.5	72.8

S2.5 Study on unknown category estimation methods

In the real-world category discovery task, the exact number of novel categories is often unknown, posing a significant challenge for model training and evaluation. Existing literature (Vaze et al., 2022a; Hao et al., 2024; Zhao et al., 2023), has explored methods to estimate the number of unknown categories (K). Building upon these studies, we conduct a comprehensive analysis to assess the effectiveness of different K -estimation methods when integrated with a stronger foundation model, specifically DINOv2 (Oquab et al., 2024).

Our experimental setup involves training three distinct GCD methods, *i.e.*, GCD (Vaze et al., 2022a), SelEx (Rastegar et al., 2024), and our proposed PartCo-SelEx (Ours) using DINOv2 pretrained weights. These trained models serve as feature extractors for the subsequent K -estimation process. We employ two off-the-shelf K -estimation techniques: GCD (Vaze et al., 2022a) and CiPR (Hao et al., 2024) K -est methods. The performance of these integrated approaches is evaluated on two fine-grained datasets: CUB (Wah et al., 2011) and Stanford-Cars (Krause et al., 2013).

 Table J. **Unknown category estimation.** Category estimation results of various K -estimation methods on different GCD methods using DINOv2.

Method	CUB		Stanford-Cars	
	GCD	CiPR	GCD	CiPR
	K -est	K -est	K -est	K -est
GCD	188	178	242	169
SelEx	219	191	229	194
PartCo-SelEx (Ours)	210	192	185	195
Ground-truth K	200		196	

Tab. J shows the estimation results of different K -estimation methods when applied to the GCD, SelEx, and PartCo-SelEx methods. The ground-truth number of categories is 200 for the CUB dataset and 196 for the Stanford-Cars dataset. We observe that the GCD K -estimation method, when paired with the GCD’s weights, significantly underestimates $K = 188$ for CUB dataset and overestimates $K = 242$ for Stanford-Cars. In contrast, the CiPR K -estimation method offers improved estimations, though still not perfectly aligned with the ground truth ($K = 178$ for CUB and $K = 169$ for Stanford-Cars).

When integrating SelEx with the K -estimation methods, the performance improves, with CiPR providing more accurate estimates ($K = 191$ for CUB and $K = 194$ for Stanford-Cars) compared to GCD’s native method ($K = 219$ for CUB and $K = 229$ for Stanford-Cars). On the other hand, our proposed PartCo-SelEx framework demonstrates the most accurate K -estimation across both datasets, achieving estimates of 210 for CUB and 185 for Stanford-Cars with the GCD K -estimation method, and $K = 192$ for CUB and $K = 195$ for Stanford-Cars with the CiPR method. These results indicate that PartCo-SelEx consistently provides K -estimates that are closer to the ground truth, particularly when using the CiPR estimation method.

 Table K. **GCD performance with estimated K .** GCD results on DINOv2 with the estimated number of categories.

Method	CUB			Stanford-Cars		
	All	Old	New	All	Old	New
GCD	68.9	77.0	65.0	62.2	72.5	57.2
SimGCD	70.4	78.1	66.7	69.7	84.8	62.3
PartCo-SimGCD (Ours)	78.8	80.3	78.0	73.9	84.9	68.6
SelEx	<u>86.1</u>	77.8	<u>90.3</u>	<u>78.3</u>	89.2	<u>73.0</u>
PartCo-SelEx (Ours)	87.6	<u>79.3</u>	91.7	80.6	<u>88.7</u>	76.8

To further evaluate the robustness of our method under challenging K -estimation scenarios, we conduct experiments using

the worst estimation method results: $K = 188$ for CUB and $K = 242$ for Stanford-Cars, as shown in Tab. K. Despite these inaccurate K -estimates, our PartCo-SelEx framework maintains superior performance compared to baseline models. Specifically, on the CUB dataset, PartCo-SelEx achieves an accuracy of 87.6% on the ‘All’ category and 91.7% on the ‘New’ category, outperforming the SelEx baseline which achieves 86.1% and 90.3% respectively. Similarly, on the Stanford-Cars dataset, PartCo-SelEx attains the highest accuracies of 80.6% for ‘All’ and 76.8% for ‘New’ categories, surpassing the SelEx baseline’s 78.3% and 73.0% respectively. These results underscore the robustness of PartCo-SelEx in handling erroneous K -estimates, ensuring consistent and reliable performance even when the estimated number of categories deviates from the ground truth.

S3 Qualitative Results

S3.1 Additional visualization of part-level correspondence labels

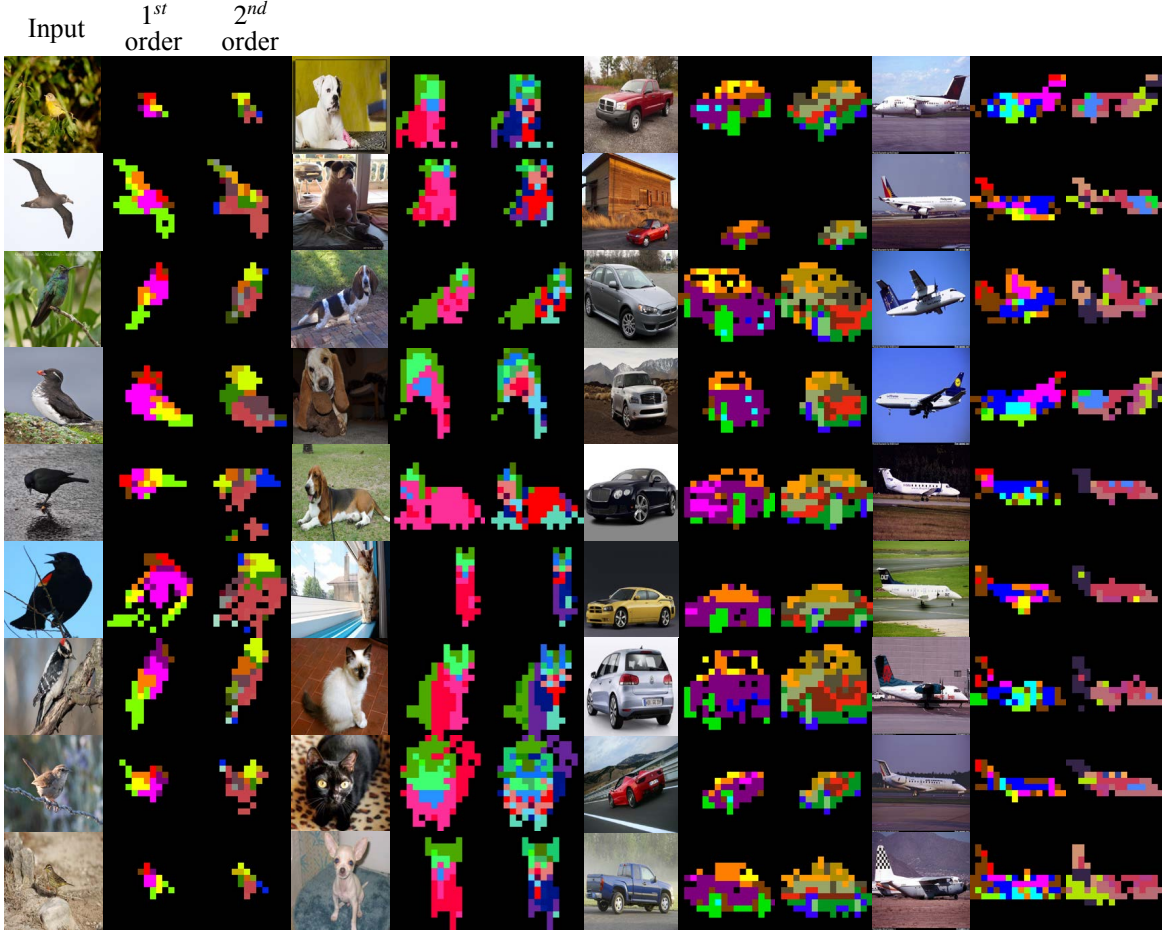


Figure B. Additional visualization of our part-level correspondence labels. For every input image, both first and second-order labels are constructed.

S3.2 Qualitative analysis of PartCo attention maps

We provide a qualitative analysis of the attention maps generated by SimGCD (Wen et al., 2023) and SelEx (Rastegar et al., 2024) when integrated with the PartCo framework (Ours), applied to the CUB (Wah et al., 2011) and Stanford-Cars (Krause et al., 2013) datasets, as shown in Fig. C & D. These attention maps originate from the final block of DINOv2 ViT backbone, utilizing a resolution of 16×16 . Following the methodology outlined in (Caron et al., 2021), we calculate the mean value across all attention heads and upsample the resulting maps to the original image resolution for visualization purposes. The analysis shows that all evaluated methods focus their attention on specific parts of the objects, effectively highlighting regions crucial for distinguishing between fine-grained categories. Notably, when combined with SimGCD, the PartCo framework (PartCo-SimGCD) tends to concentrate on particular parts of the object, such as the wings of a bird in the CUB dataset or the wheels of a car in the Stanford-Cars dataset. This targeted focus underscores PartCo-SimGCD’s ability to hone in on key discriminative features essential for accurate category differentiation. In contrast, integrating PartCo with SelEx (PartCo-SelEx) results in attention maps that cover a broader range of object parts, capturing multiple fine-grained details simultaneously. For example, PartCo-SelEx not only highlights the wings but also the body and head of the bird in the CUB dataset, and the wheels, doors, and headlights of the car in the Stanford-Cars dataset.

These observations indicate that while both PartCo-SimGCD and PartCo-SelEx effectively utilize part-level information to enhance attention mechanisms, PartCo-SelEx exhibits a more comprehensive focus on multiple object parts. This broader attention coverage can potentially lead to a more nuanced understanding of fine-grained categories, thereby improving

the model’s ability to generalize across diverse and complex datasets. Overall, the qualitative analysis demonstrates the robustness and effectiveness of the PartCo framework in refining attention maps, highlighting its superior capability to capture meaningful visual regions.

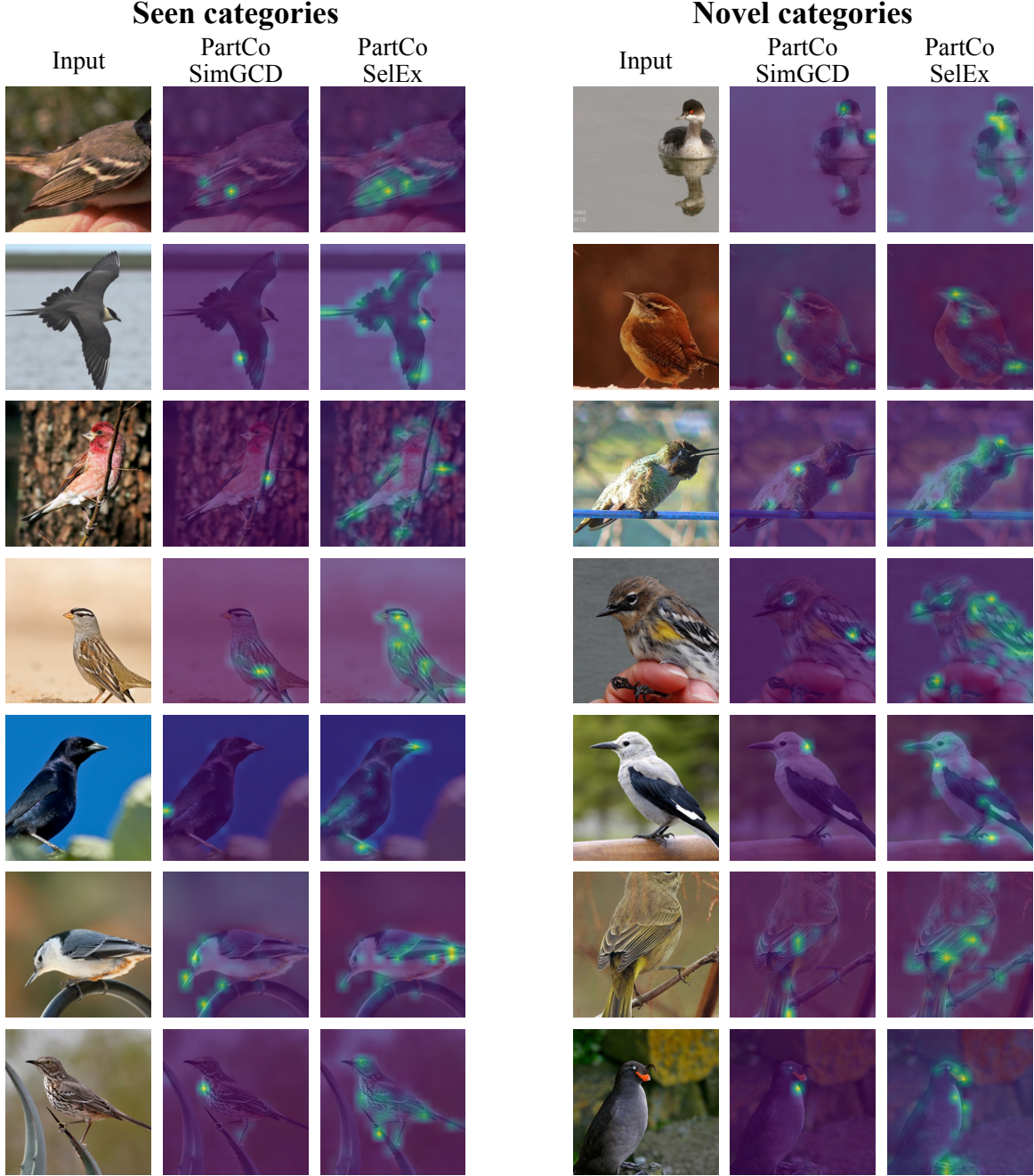


Figure C. Attention maps on CUB dataset. Visualization of attention maps generated by the PartCo framework integrated with SimGCD and SelEx for the CUB dataset. The PartCo-SimGCD model highlights specific regions such as the wings and head of the bird, indicating focused attention on key discriminative features. In contrast, the PartCo-SelEx model displays a broader attention distribution, encompassing multiple parts including the wings, body, and tail.



Figure D. Attention maps on Stanford-Cars dataset. Visualization of attention maps generated by the PartCo framework integrated with SimGCD and SelEx for the Stanford-Cars dataset. The PartCo-SimGCD model concentrates on distinct parts like the wheels and headlights of the car, demonstrating targeted attention on essential distinguishing features. Meanwhile, the PartCo-SelEx model exhibits a wider area of focus, covering various components such as the wheels, doors, and overall body structure.

S4 Success & Failure Case Analysis

Tab. L reports the per-class accuracy difference between PartCo and the Baseline on CUB, highlighting the 10 classes (see Fig. E) with the largest gains (“Success”) and the 10 classes (see Fig. F) with the largest drops (“Failure”). We observe that the dominant pattern in the success cases is that PartCo helps on fine-grained, part-dependent species, while the failures largely correspond to classes where global shape or coarse appearance is more informative than local details.

Success cases (part-dependent classes). The classes where PartCo substantially outperforms the Baseline (e.g., *Magnolia Warbler*, *Marsh Wren*, *Black-throated Blue Warbler*, *Seaside Sparrow*, *Barn Swallow*) are mostly small passerines with very similar global silhouettes but distinctive, localized plumage. These species are hard to distinguish using only a global descriptor because they all look like “small round songbirds” in terms of overall shape. Instead, they are identified by specific local cues: for instance, *Magnolia Warbler* (ID: 169) has a characteristic tail and yellow underparts, while *Black-throated Blue Warbler* (ID: 160) is defined by a dark throat patch and contrasting flanks. By explicitly learning and matching part-level features, PartCo can anchor its predictions on these discriminative regions, turning many near-0% Baseline accuracies into high per-class accuracy. Notably, several of these gains occur on unseen classes, indicating that part-level cues provide a transferable signal that generalizes beyond the seen taxonomy.

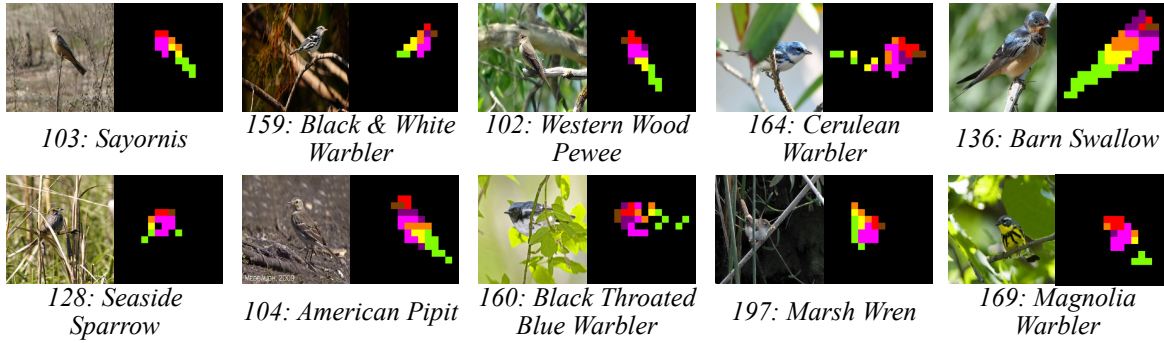


Figure E. **Visualization of the top 10 success cases on the CUB dataset.** We display sample images for each class alongside their corresponding part-level correspondence labels.

Table L. **Per-class success and failure analysis.** We report the 10 classes with the largest accuracy gains (*Success*, top) and the 10 classes with the largest drops (*Failure*, bottom) when replacing the Baseline with PartCo, along with their seen/unseen type labels. Δ denotes the ACC difference.

ID	Class Name	Type category	ACC		Δ
			Baseline	+PartCo	
Success Cases (top 10)					
169	Magnolia Warbler	Seen	0.0	100.0	+100.0
197	Marsh Wren	Seen	3.3	100.0	+96.7
160	Black-throated Blue Warbler	Unseen	0.0	76.7	+76.7
164	Cerulean Warbler	Seen	0.0	75.0	+75.0
128	Seaside Sparrow	Unseen	0.0	73.3	+73.3
136	Barn Swallow	Seen	21.3	93.3	+72.0
104	American Pipit	Unseen	6.7	70.0	+63.3
102	Western Wood Pewee	Unseen	40.0	90.0	+50.0
159	Black & White Warbler	Seen	46.7	96.7	+50.0
103	Sayornis	Unseen	53.3	100.0	+46.7
Failure Cases (top 10)					
129	Song Sparrow	Unseen	76.7	41.7	-35.0
89	Hooded Merganser	Seen	63.3	33.3	-30.0
162	Canada Warbler	Seen	83.3	58.3	-25.0
134	Cape Glossy Starling	Seen	100.0	75.0	-25.0
110	Geococcyx	Unseen	73.3	50.0	-23.3
180	Wilson’s Warbler	Seen	96.7	75.0	-21.7
120	Fox Sparrow	Seen	80.0	60.0	-20.0
130	Tree Sparrow	Seen	96.7	76.7	-20.0
32	Mangrove Cuckoo	Seen	94.7	75.5	-19.2
184	Louisiana Waterthrush	Unseen	83.3	65.0	-18.3

Failure cases (global/shape-biased or ambiguous classes). The main failures (*e.g.*, *Song Sparrow*, *Hooded Merganser*, *Mangrove Cuckoo*, *Wilson’s Warbler*) tend to be classes where (i) global structure is more distinctive than any single local patch, or (ii) local patches are visually ambiguous across species. For instance, *Mangrove Cuckoo* (ID: 32) has a very characteristic elongated body and tail that are easily captured by a global representation, while its local textures (brown and grey feathers) are relatively generic. Similarly, *Wilson’s Warbler* (ID: 180) is almost uniformly yellow with a small dark cap; zoomed-in patches mostly look like “yellow feathers”, which are hard to separate from other yellow warblers. In such cases, forcing the model to over-emphasize parts can down-weight useful global shape cues and make the decision boundary noisier. For *Hooded Merganser* (ID: 89), qualitative inspection of CUB samples, as shown in Fig. G, suggests substantial appearance variation between juveniles and adults, which further increases intra-class variability at the part level and is not explicitly modeled in our current design.

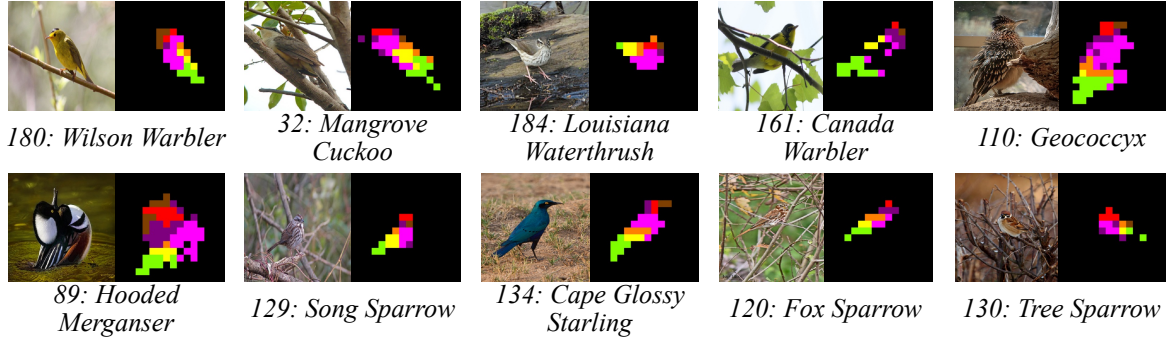


Figure F. **Visualization of the top 10 failure cases on the CUB dataset.** We display sample images for each class alongside their corresponding part-level correspondence labels.



Figure G. **Example Hooded Merganser (ID 89) images from CUB**, showing large intra-class variation (juvenile vs. adult, biological development). This variation makes part-level cues ambiguous and contributes to PartCo’s reduced accuracy on this class compared to the Baseline.

Overall, however, the magnitude of the largest drops is clearly smaller than the strongest gains (*e.g.*, maximum decrease of around 35 points vs. gains up to 100 points), and several of the biggest improvements occur on unseen classes. This supports our main claim that part-level cues are an effective vehicle for transferring fine-grained discriminative knowledge from seen to unseen categories, while suggesting that future work could mitigate the residual failures by better handling intra-class appearance factors such as age and biological species-specific development.

S5 Limitations & Future Work

The current PartCo framework relies on foundation models that provide spatially indexed patch token representations, as in recent transformer-based architectures. This makes PartCo less directly applicable to backbones without an explicit patch-token structure, such as many convolutional networks (CNN) or older vision architectures. In addition, PartCo deliberately keeps the part-label construction stage lightweight and decoupled from the host GCD training loop by using a PCA-based pipeline on frozen patch features. While this design preserves the plug-in nature of the framework, it does not explore more structured learned region decomposition methods. A natural direction for future work is to replace the current PCA-based construction with slot-based (Locatello et al., 2020) or other object-centric decomposition approaches while keeping the PartCo correspondence loss fixed, and to test whether more structured region assignments can further improve GCD. Moreover, because PartCo’s supervisory signal is currently defined on the native ViT patch grid, the spatial resolution of correspondence is inherently limited by the number of patch tokens. Another promising direction is to densify this signal using feature upsampling methods (Fu et al., 2024; Wimmer et al., 2026), so that part-level correspondence can operate on finer spatial representations without abandoning the pretrained ViT backbone.